

```
%root_path = fileparts(mfilename("fullpath"));
%root_path = pwd;

% retrieve selected cuts used for further fitting
[HN_FFCutsFits,CutFitFileNG] =
recover_selected_cuts('HN_FFCutsFits','e:\SHARE\Fe\Data\analysis\07_fit_with_multic
ut_allJ\sel_cuts');
```

```
fn = fieldnames(HN_FFCutsFits);
nf = 11
```

```
nf =
11
```

```
%for i = 11:numel(fn)
%selectCuts = fn{nf};
selectCuts = 'Ei800HN_offH2H0minFF'
```

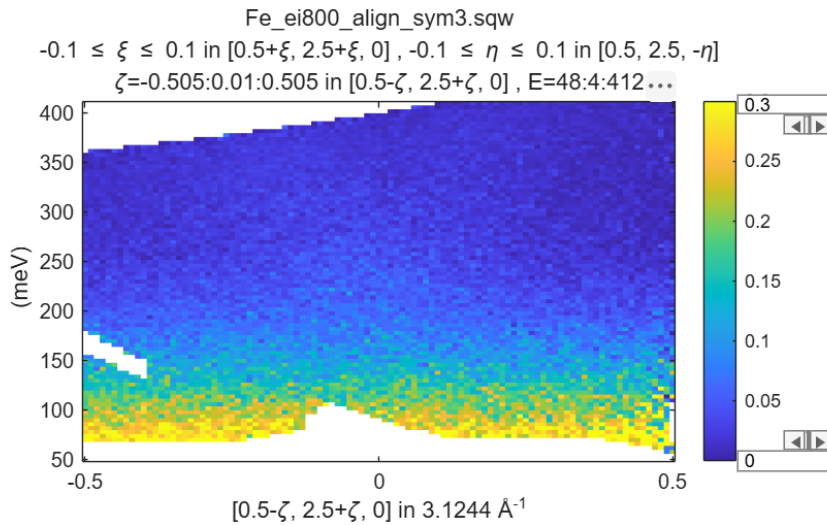
```
selectCuts =
'Ei800HN_offH2H0minFF'
```

```
if ismember(selectCuts(1:3),{'msk','dat'})||...
    contains(selectCuts,'all_fit_par')
    %continue
    error('Can not fit field: %s',selectCuts);
end
```

```
fprintf('ProcessiHN cuts: %s \n', selectCuts);
```

```
ProcessiHN cuts: Ei800HN_offH2H0minFF
```

```
cuts_list = HN_FFCutsFits.(selectCuts);
if ~iscell(cuts_list)
    cuts_list = {cuts_list};
end
if contains(selectCuts,'Ei200')
    min_Ei = 30;
    mi = maps_instrument(200,600,'S');
end
if contains(selectCuts,'Ei400')
    min_Ei = 40;
    mi = maps_instrument(401,600,'S');
end
if contains(selectCuts,'Ei800')
    min_Ei = 70;
    mi = maps_instrument(786,600,'S');
end
sample=IX_sample(true,[1,0,0],[0,1,0],'cuboid',[0.04,0.03,0.02]);
for j=1:numel(cuts_list)
    cuts_list{j} = cuts_list{j}.set_instrument(mi);
    cuts_list{j} = cuts_list{j}.set_sample(sample);
    plot(cuts_list{j}); lz 0 0.3; keep_figure;
end
```



```
dir_name = "HN";
```

```
dE_step = 2; %original energy transfer step data were binned to. No point in going finer
```

```
half_dE = 5; % half width of data binning
```

```
en_mm = cuts_list{1}.data.img_range(:,4);
```

```
en_mm(1) = max(min_Ei,en_mm(1));
```

```
en_mm(1) = 220;
```

```
%en_mm(2) = 380;
```

```
cut_en = en_mm(1)+half_dE:2*half_dE:(en_mm(2)-half_dE);
```

```
cut_dimensions = 1;
```

```
HN_FFcutsFits.(selectCuts) = cuts_list;
```

```
HN_FFcutsFits = fit_multicutsFF_along_direction(...
```

```
HN_FFcutsFits,selectCuts,cut_dimensions,cut_en,dE_step,half_dE,true);
```

```
*****
**** En = 225±5 going to 405
*****
```

```
-----
Beginning fit (max 20 iterations)-----
```

```
Starting point
```

```
-----
Total time = 0.81104s    Reduced Chi^2 = 2.6429
```

```
Free parameter values:
```

```
10      0.7917      250      34      0.04194
```

```
-----
Iteration = 1
```

```
-----
Total time = 4.1097s    Reduced Chi^2 = 10.0643
```

```
Free parameter values:
```

```
400.6    22.87    -1.118e+04    663.1    0.0291
```

```
Total time = 4.6795s    Reduced Chi^2 = 2.6881
```

```
Free parameter values:
```

```
174.4    11.05    -5108    191.5    0.03525
```

```
Total time = 5.2516s    Reduced Chi^2 = 10.1708
```

```
Free parameter values:
```

```
66.54    4.401    -1592    78.26    0.03964
```

```

Total time = 5.831s    Reduced Chi^2 = 2.4679    Levenberg-Marquardt = 1000
Free parameter values:
    15.76    1.161    63.22    38.28    0.04171
-----
Iteration = 2
-----
Total time = 8.6674s    Reduced Chi^2 = 2.2125    Levenberg-Marquardt = 100
Free parameter values:
    19    1.386    52.1    36.57    0.04098
-----
Iteration = 3
-----
Total time = 11.5276s    Reduced Chi^2 = 1.6914    Levenberg-Marquardt = 10
Free parameter values:
    23.64    1.738    43.35    34.76    0.03885
-----
Iteration = 4
-----
Total time = 14.4258s    Reduced Chi^2 = 1.1238    Levenberg-Marquardt = 1
Free parameter values:
    30.42    2.141    39.22    34.94    0.03417
-----
Iteration = 5
-----
Total time = 17.2034s    Reduced Chi^2 = 0.98621    Levenberg-Marquardt = 0.1
Free parameter values:
    43.94    2.781    47.24    36.02    0.03038
-----
Iteration = 6
-----
Total time = 19.9702s    Reduced Chi^2 = 0.97765    Levenberg-Marquardt = 0.01
Free parameter values:
    78.36    4.09    25.59    45.52    0.02859
-----
Iteration = 7
-----
Total time = 22.645s    Reduced Chi^2 = 0.98216    Levenberg-Marquardt = 0.001
Free parameter values:
    149.9    7.413    -61.63    72.69    0.02834
Total time = 23.2269s    Reduced Chi^2 = 0.96644    Levenberg-Marquardt = 0.01
Free parameter values:
    108.7    5.182    2.389    54.16    0.02848
-----
Iteration = 8
-----
Total time = 26.0394s    Reduced Chi^2 = 0.98343    Levenberg-Marquardt = 0.001
Free parameter values:
    211.3    6.113    -106.9    80.12    0.0286
Total time = 26.6024s    Reduced Chi^2 = 0.96669    Levenberg-Marquardt = 0.01
Free parameter values:
    145.1    5.405    -32.1    62.38    0.02852
Total time = 27.1494s    Reduced Chi^2 = 0.96542    Levenberg-Marquardt = 0.1
Free parameter values:
    121.2    5.261    -8.416    56.73    0.02853
-----
Iteration = 9
-----
Total time = 29.9167s    Reduced Chi^2 = 0.96514    Levenberg-Marquardt = 0.01
Free parameter values:
    161.2    5.269    -47.24    65.12    0.02856
-----
Fit converged:

Parameter values (free parameters with error estimates):

```

Foreground parameter values:

1	1		
2	8		
3	161.2	+/-	1595
4	5.269	+/-	57.11
5	-47.24	+/-	1436
6	65.12	+/-	482.6
7	0		
8	0		
9	0		
10	0		

Background parameter values:

1	0.02856	+/-	0.002077
2	0		
3	0		

Covariance matrix for free parameters:

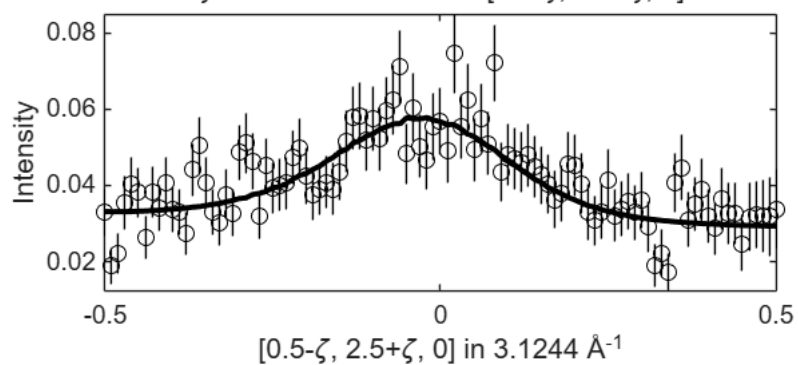
1.0000	0.9457	-0.9966	0.9924	-0.2898
0.9457	1.0000	-0.9321	0.9774	-0.4126
-0.9966	-0.9321	1.0000	-0.9874	0.2236
0.9924	0.9774	-0.9874	1.0000	-0.3169
-0.2898	-0.4126	0.2236	-0.3169	1.0000

Fe_ei800_align_sym3.sqw

S= 5.3± 57; J0= 65±4.8e+02; J1= 0± 0; gamma=161.226±1595.0

$\xi \leq 0.1$ in $[0.5+\xi, 2.5+\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[0.5, 2.5, -\eta]$, $220 \leq$

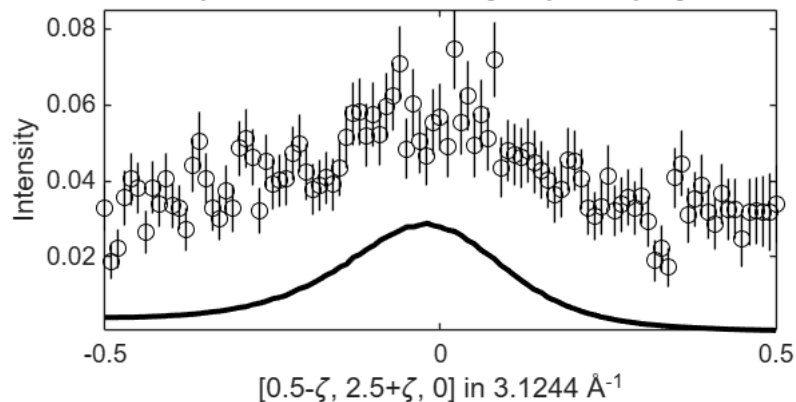
$\zeta = -0.505:0.01:0.505$ in $[0.5-\zeta, 2.5+\zeta, 0]$



Fe_ei800_align_sym3.sqw

$\xi \leq 0.1$ in $[0.5+\xi, 2.5+\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[0.5, 2.5, -\eta]$, $220 \leq$

$\zeta = -0.505:0.01:0.505$ in $[0.5-\zeta, 2.5+\zeta, 0]$



**** En = 235±5 going to 405

Beginning fit (max 20 iterations)-----

Starting point

Total time = 0.57602s Reduced Chi^2 = 5.3691

Free parameter values:

161.2	5.269	-47.24	65.12	0.03762
-------	-------	--------	-------	---------

Iteration = 1

Total time = 3.4004s Reduced Chi^2 = 0.97747

Levenberg-Marquardt = 0.1

Free parameter values:

127.2	6.006	-44.9	66.86	0.02563
-------	-------	-------	-------	---------

Iteration = 2

Total time = 6.229s Reduced Chi^2 = 0.97596

Levenberg-Marquardt = 0.01

Free parameter values:

115.5	8.997	-6.222	66.91	0.02384
-------	-------	--------	-------	---------

Iteration = 3

Total time = 9.0426s Reduced Chi^2 = 2.2475

Levenberg-Marquardt = 0.001

Free parameter values:

33.08	16.5	55.17	61.41	0.02343
-------	------	-------	-------	---------

Total time = 9.6414s Reduced Chi^2 = 0.96155

Levenberg-Marquardt = 0.01

Free parameter values:

91.95	11.69	14.26	64.84	0.02361
-------	-------	-------	-------	---------

Iteration = 4

Total time = 12.4748s Reduced Chi^2 = 4.467

Levenberg-Marquardt = 0.001

Free parameter values:

4.558	22.16	68.61	57.45	0.02335
-------	-------	-------	-------	---------

Total time = 13.0578s Reduced Chi^2 = 0.98717

Levenberg-Marquardt = 0.01

Free parameter values:

64.3	15.33	31.27	62.25	0.02351
------	-------	-------	-------	---------

Total time = 13.6368s Reduced Chi^2 = 0.94706

Levenberg-Marquardt = 0.1

Free parameter values:

84.19	12.96	18.19	63.83	0.02359
-------	-------	-------	-------	---------

Iteration = 5

```

-----
Total time = 16.3636s    Reduced Chi^2 = 1.0648    Levenberg-Marquardt = 0.01
Free parameter values:
    54.24    19.18    36.61    63.97    0.02341
Total time = 16.9473s    Reduced Chi^2 = 0.94972    Levenberg-Marquardt = 0.1
Free parameter values:
    74.98    14.97    24.3    63.85    0.02353
Total time = 17.5029s    Reduced Chi^2 = 0.94233    Levenberg-Marquardt = 1
Free parameter values:
    81.38    13.61    19.99    63.82    0.02358
-----
Iteration = 6
-----
Total time = 20.4018s    Reduced Chi^2 = 0.94679    Levenberg-Marquardt = 0.1
Free parameter values:
    68.27    14.9    27.37    61.84    0.02354
Total time = 20.9499s    Reduced Chi^2 = 0.95209    Levenberg-Marquardt = 1
Free parameter values:
    77.3    14.03    22.15    63.18    0.02356
Total time = 21.5634s    Reduced Chi^2 = 0.95019    Levenberg-Marquardt = 10
Free parameter values:
    80.12    13.75    20.63    63.61    0.02358
Total time = 22.1484s    Reduced Chi^2 = 0.9467    Levenberg-Marquardt = 1000
Free parameter values:
    81.26    13.63    20.06    63.8    0.02358
Total time = 22.7054s    Reduced Chi^2 = 0.94261    Levenberg-Marquardt = 100000
Free parameter values:
    81.37    13.61    20    63.82    0.02358
Total time = 23.2625s    Reduced Chi^2 = 0.94978    Levenberg-Marquardt = 10000000
Free parameter values:
    81.38    13.61    19.99    63.82    0.02358
*** No improvement in chi-squared over previous iteration ***
Total time = 23.2703s    Reduced Chi^2 = 0.94233
Free parameter values:
    81.38    13.61    19.99    63.82    0.02358
-----

```

Fit converged:

Parameter values (free parameters with error estimates):

Foreground parameter values:

```

-----
1          1
2          8
3          81.38 +/- 141.8
4          13.61 +/- 80.89
5          19.99 +/- 133.2
6          63.82 +/- 105.3
7          0
8          0
9          0
10         0

```

Background parameter values:

```

-----
1          0.02358 +/- 0.001703
2          0
3          0

```

Covariance matrix for free parameters:

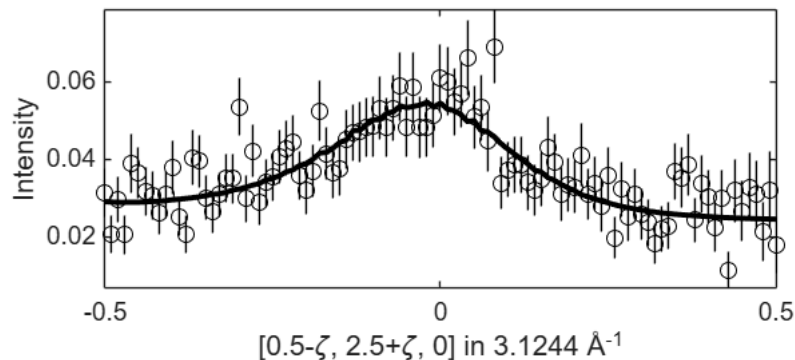
1.0000	-0.6135	0.3871	-0.4041	-0.1213
-0.6135	1.0000	-0.9656	0.9702	-0.5549
0.3871	-0.9656	1.0000	-0.9998	0.6824
-0.4041	0.9702	-0.9998	1.0000	-0.6733
-0.1213	-0.5549	0.6824	-0.6733	1.0000

Fe_ei800_align_sym3.sqw

S= 14± 81; J0= 64±1.1e+02; J1= 0± 0; gamma=81.3849±141.77

$\xi \leq 0.1$ in $[0.5+\xi, 2.5+\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[0.5, 2.5, -\eta]$, $230 \leq$

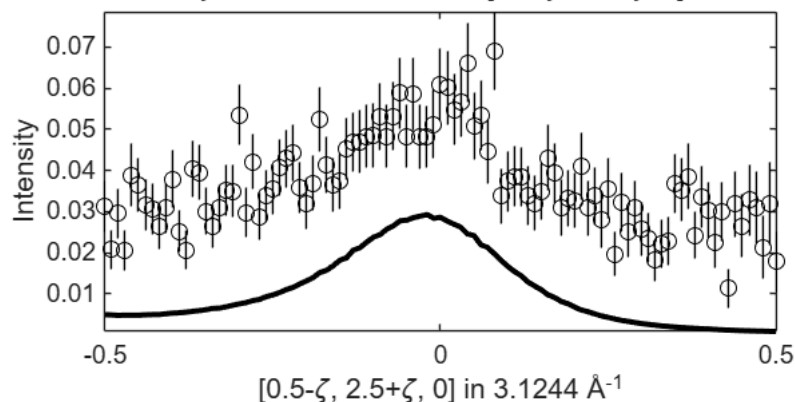
$\zeta = -0.505:0.01:0.505$ in $[0.5-\zeta, 2.5+\zeta, 0]$



Fe_ei800_align_sym3.sqw

$\xi \leq 0.1$ in $[0.5+\xi, 2.5+\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[0.5, 2.5, -\eta]$, $230 \leq$

$\zeta = -0.505:0.01:0.505$ in $[0.5-\zeta, 2.5+\zeta, 0]$



**** En = 245±5 going to 405

Beginning fit (max 20 iterations)-----

Starting point

Total time = 0.56366s Reduced Chi^2 = 6.9217

Free parameter values:

81.38	13.61	19.99	63.82	0.0335
-------	-------	-------	-------	--------

Iteration = 1

Total time = 3.422s Reduced Chi^2 = 0.96744 Levenberg-Marquardt = 0.1

Free parameter values:

77.71	14.01	32.5	65.4	0.02127
-------	-------	------	------	---------

Iteration = 2

Total time = 6.2807s Reduced Chi^2 = 0.93395 Levenberg-Marquardt = 0.01

Free parameter values:

95.16	15.96	77.67	64.47	0.01926
-------	-------	-------	-------	---------

Iteration = 3

Total time = 9.1527s	Reduced Chi^2 = 1.2836	Levenberg-Marquardt = 0.001
Free parameter values:		
55.35	27.58	118.3
65.53	0.01865	
Total time = 9.7501s	Reduced Chi^2 = 0.94197	Levenberg-Marquardt = 0.01
Free parameter values:		
86.4	20.1	101.3
64.5	0.01884	
Total time = 10.3257s	Reduced Chi^2 = 0.93461	Levenberg-Marquardt = 0.1
Free parameter values:		
93.27	17.36	86.83
64.4	0.01907	
Total time = 10.8939s	Reduced Chi^2 = 0.92813	Levenberg-Marquardt = 10
Free parameter values:		
95.06	16.11	78.48
64.45	0.01924	

Iteration = 4

Total time = 13.7114s	Reduced Chi^2 = 0.93202	Levenberg-Marquardt = 1
Free parameter values:		
92.83	16.6	82.01
64.05	0.01918	
Total time = 14.327s	Reduced Chi^2 = 0.93284	Levenberg-Marquardt = 10
Free parameter values:		
94.38	16.27	79.57
64.32	0.01922	
Total time = 14.9189s	Reduced Chi^2 = 0.93191	Levenberg-Marquardt = 100
Free parameter values:		
94.85	16.16	78.82
64.41	0.01924	
Total time = 15.502s	Reduced Chi^2 = 0.93574	Levenberg-Marquardt = 10000
Free parameter values:		
95.04	16.12	78.52
64.45	0.01924	
Total time = 16.0861s	Reduced Chi^2 = 0.93179	Levenberg-Marquardt = 1000000
Free parameter values:		
95.06	16.11	78.49
64.45	0.01924	
Total time = 16.7089s	Reduced Chi^2 = 0.93252	Levenberg-Marquardt = 100000000
Free parameter values:		
95.06	16.11	78.48
64.45	0.01924	
*** No improvement in chi-squared over previous iteration ***		
Total time = 16.7227s	Reduced Chi^2 = 0.92813	
Free parameter values:		
95.06	16.11	78.48
64.45	0.01924	

Fit converged:

Parameter values (free parameters with error estimates):

Foreground parameter values:

1	1		
2	8		
3	95.06	+/-	483.8
4	16.11	+/-	53.58
5	78.48	+/-	237
6	64.45	+/-	63.9
7	0		
8	0		
9	0		
10	0		

Background parameter values:

1	0.01924	+/-	0.001554
2	0		

Covariance matrix for free parameters:

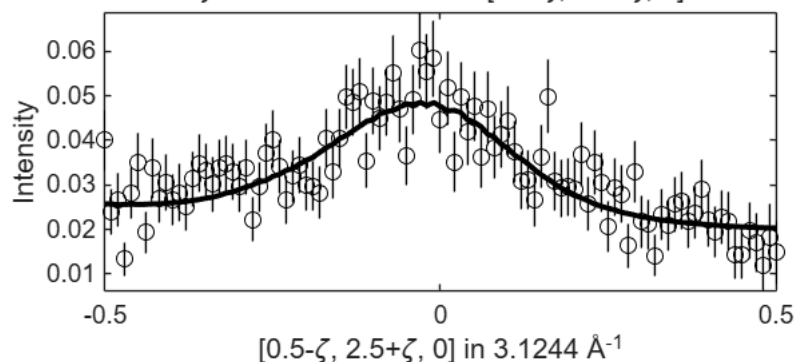
```
-----
  1.0000   -0.9760   -0.9181   0.9057   0.3804
 -0.9760   1.0000   0.8099   -0.7919   -0.2360
 -0.9181   0.8099   1.0000   -0.9995   -0.6145
  0.9057   -0.7919   -0.9995   1.0000   0.6291
  0.3804   -0.2360   -0.6145   0.6291   1.0000
-----
```

Fe_ei800_align_sym3.sqw

S= 16± 54; J0= 64± 64; J1= 0± 0; gamma=95.0594±483.802

$\xi \leq 0.1$ in $[0.5+\xi, 2.5+\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[0.5, 2.5, -\eta]$, $240 \leq$

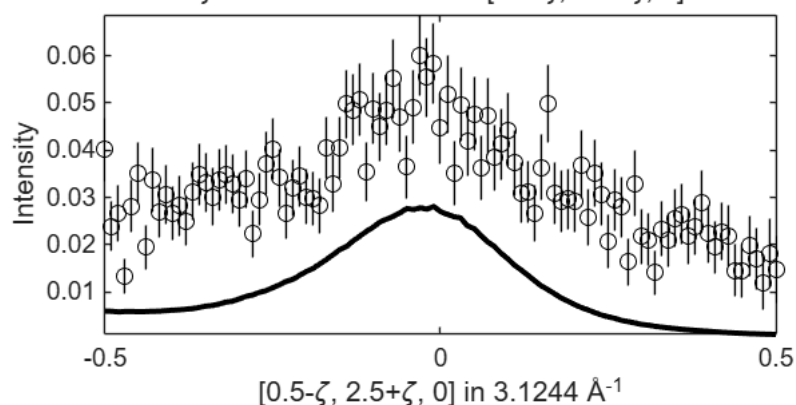
$\zeta=-0.505:0.01:0.505$ in $[0.5-\zeta, 2.5+\zeta, 0]$



Fe_ei800_align_sym3.sqw

$\xi \leq 0.1$ in $[0.5+\xi, 2.5+\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[0.5, 2.5, -\eta]$, $240 \leq$

$\zeta=-0.505:0.01:0.505$ in $[0.5-\zeta, 2.5+\zeta, 0]$



**** En = 255±5 going to 405

Beginning fit (max 20 iterations)-----

Starting point

Total time = 0.62962s Reduced Chi^2 = 8.3037

Free parameter values:

95.06 16.11 78.48 64.45 0.03008

Iteration = 1

Total time = 3.5036s Reduced Chi^2 = 1.0724 Levenberg-Marquardt = 0.1

Free parameter values:

101.5 14.91 60.58 68.22 0.01664

Iteration = 2

```
-----
Total time = 6.4378s   Reduced Chi^2 = 1.0844   Levenberg-Marquardt = 0.01
Free parameter values:
    163.5           14.16           40.28           79.29           0.01492
Total time = 7.0215s   Reduced Chi^2 = 1.0406   Levenberg-Marquardt = 0.1
Free parameter values:
    122.6           14.85           52.17           71.46           0.01523
-----
```

Iteration = 3

```
-----
Total time = 9.8802s   Reduced Chi^2 = 1.1654   Levenberg-Marquardt = 0.01
Free parameter values:
    204.3           11.46           27.78           82.03           0.01488
Total time = 10.5119s   Reduced Chi^2 = 1.0445   Levenberg-Marquardt = 0.1
Free parameter values:
    149.6           13.86           45.15           74.7           0.01501
Total time = 11.0802s   Reduced Chi^2 = 1.0354   Levenberg-Marquardt = 1
Free parameter values:
    131.4           14.57           49.52           72.44           0.01513
-----
```

Iteration = 4

```
-----
Total time = 14.01s   Reduced Chi^2 = 1.0404   Levenberg-Marquardt = 0.1
Free parameter values:
    158.2           13.53           42.01           75.58           0.01498
Total time = 14.631s   Reduced Chi^2 = 1.0396   Levenberg-Marquardt = 1
Free parameter values:
    140.1           14.26           46.99           73.41           0.01506
Total time = 15.2442s   Reduced Chi^2 = 1.0378   Levenberg-Marquardt = 10
Free parameter values:
    134.2           14.48           48.67           72.74           0.01511
Total time = 15.8475s   Reduced Chi^2 = 1.0401   Levenberg-Marquardt = 1000
Free parameter values:
    131.7           14.56           49.43           72.47           0.01513
Total time = 16.4565s   Reduced Chi^2 = 1.0377   Levenberg-Marquardt = 100000
Free parameter values:
    131.5           14.57           49.51           72.44           0.01513
Total time = 17.0072s   Reduced Chi^2 = 1.0405   Levenberg-Marquardt = 10000000
Free parameter values:
    131.4           14.57           49.52           72.44           0.01513
*** No improvement in chi-squared over previous iteration ***
Total time = 17.0156s   Reduced Chi^2 = 1.0354
Free parameter values:
    131.4           14.57           49.52           72.44           0.01513
-----
```

Fit converged:

Parameter values (free parameters with error estimates):

Foreground parameter values:

```
-----
1           1
2           8
3           131.4 +/- 779
4           14.57 +/- 25.29
5           49.52 +/- 478.5
6           72.44 +/- 199.4
7           0
8           0
9           0
10          0
```

Background parameter values:

```
-----
1      0.01513   +/-   0.001429
2              0
3              0
```

Covariance matrix for free parameters:

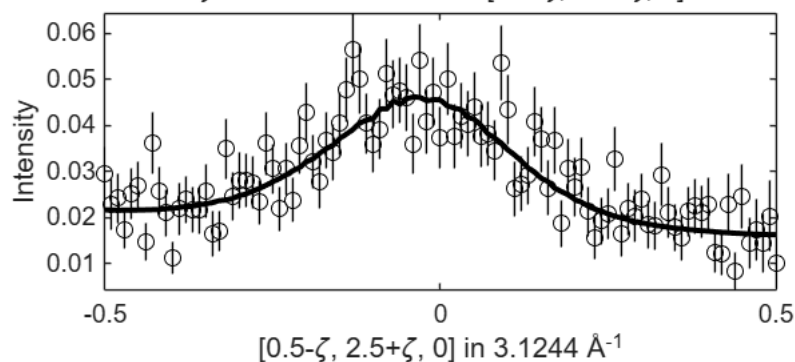
```
-----
1.0000   0.4373  -0.9711   0.9686   0.3388
0.4373   1.0000  -0.6393   0.6470   0.7313
-0.9711 -0.6393   1.0000  -0.9999  -0.4900
0.9686   0.6470  -0.9999   1.0000   0.4952
0.3388   0.7313  -0.4900   0.4952   1.0000
```

Fe_ei800_align_sym3.sqw

S= 15± 25; J0= 72±2e+02; J1= 0± 0; gamma=131.44±779.033

$\xi \leq 0.1$ in $[0.5+\xi, 2.5+\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[0.5, 2.5, -\eta]$, $250 \leq$

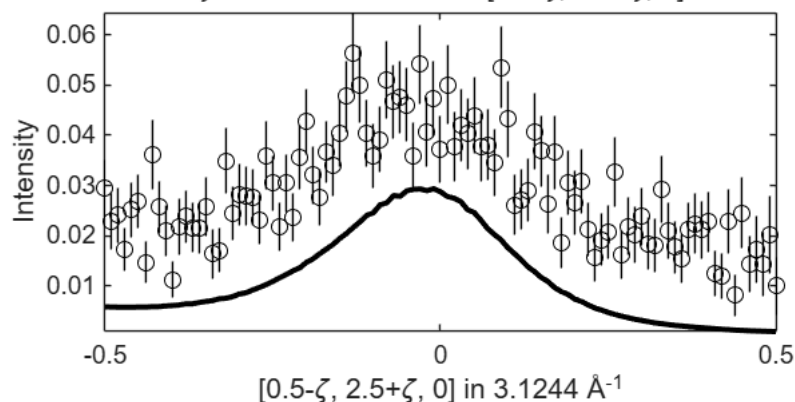
$\zeta = -0.505:0.01:0.505$ in $[0.5-\zeta, 2.5+\zeta, 0]$



Fe_ei800_align_sym3.sqw

$\xi \leq 0.1$ in $[0.5+\xi, 2.5+\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[0.5, 2.5, -\eta]$, $250 \leq$

$\zeta = -0.505:0.01:0.505$ in $[0.5-\zeta, 2.5+\zeta, 0]$



**** En = 265±5 going to 405

Beginning fit (max 20 iterations)-----

Starting point

Total time = 0.61394s Reduced Chi^2 = 9.6014

Free parameter values:

131.4 14.57 49.52 72.44 0.02778

Iteration = 1

```

-----
Total time = 3.5905s    Reduced Chi^2 = 0.95438    Levenberg-Marquardt = 0.1
Free parameter values:
    141.4      14.94      53.95      77.19      0.01446
-----
Iteration = 2
-----
Total time = 6.6293s    Reduced Chi^2 = 0.93266    Levenberg-Marquardt = 0.01
Free parameter values:
    218.5      20.43      77.48      92.63      0.01234
-----
Iteration = 3
-----
Total time = 9.6707s    Reduced Chi^2 = 1.1429    Levenberg-Marquardt = 0.001
Free parameter values:
    548.8      32.54      43.03      156.6      0.01203
Total time = 10.2748s    Reduced Chi^2 = 0.92934    Levenberg-Marquardt = 0.01
Free parameter values:
    335        24.75      82.98      112        0.01215
-----
Iteration = 4
-----
Total time = 13.294s    Reduced Chi^2 = 1.2103    Levenberg-Marquardt = 0.001
Free parameter values:
    833.2      30.79      41.5      177.8      0.01204
Total time = 13.9152s    Reduced Chi^2 = 0.9292    Levenberg-Marquardt = 0.01
Free parameter values:
    503        26.97      78.17      132.1      0.01209
-----
Fit converged:

Parameter values (free parameters with error estimates):
-----

Foreground parameter values:
-----
    1          1
    2          8
    3        503 +/- 2.316e+04
    4        26.97 +/- 1188
    5        78.17 +/- 560.2
    6        132.1 +/- 4042
    7          0
    8          0
    9          0
   10          0

Background parameter values:
-----
    1        0.01209 +/- 0.001419
    2          0
    3          0

Covariance matrix for free parameters:
-----
    1.0000    0.9822   -0.6929    0.9964   -0.4253
    0.9822    1.0000   -0.5521    0.9945   -0.4587
   -0.6929   -0.5521    1.0000   -0.6336    0.0017
    0.9964    0.9945   -0.6336    1.0000   -0.4391
   -0.4253   -0.4587    0.0017   -0.4391    1.0000

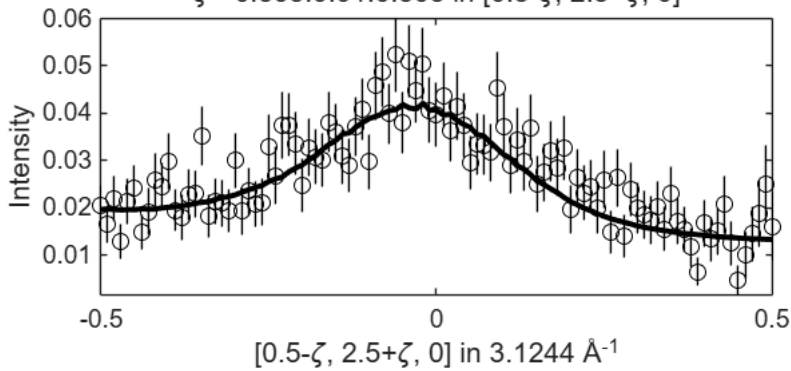
```

Fe_ei800_align_sym3.sqw

: 27±1.2e+03; J0=1.3e+02±4e+03; J1= 0± 0; gamma=503.033±231

$\xi \leq 0.1$ in $[0.5+\xi, 2.5+\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[0.5, 2.5, -\eta]$, $260 \leq$

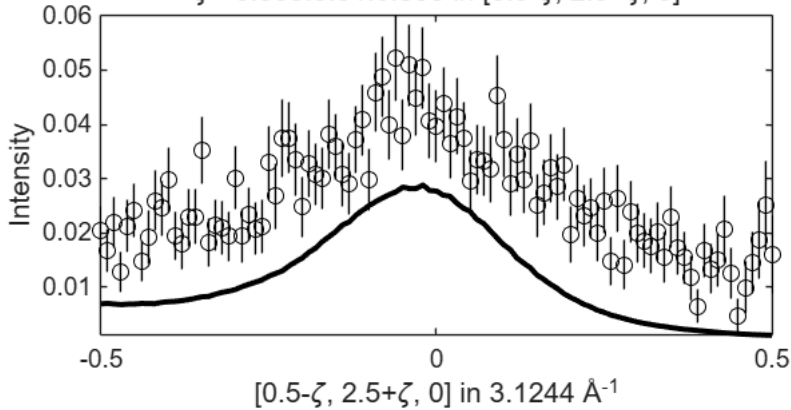
$\zeta=-0.505:0.01:0.505$ in $[0.5-\zeta, 2.5+\zeta, 0]$



Fe_ei800_align_sym3.sqw

$\xi \leq 0.1$ in $[0.5+\xi, 2.5+\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[0.5, 2.5, -\eta]$, $260 \leq$

$\zeta=-0.505:0.01:0.505$ in $[0.5-\zeta, 2.5+\zeta, 0]$



**** En = 275±5 going to 405

Beginning fit (max 20 iterations)-----

Starting point

Total time = 0.64195s Reduced Chi^2 = 10.3108

Free parameter values:

503	26.97	78.17	132.1	0.02581
-----	-------	-------	-------	---------

Iteration = 1

Total time = 3.7606s Reduced Chi^2 = 1.0926 Levenberg-Marquardt = 0.1

Free parameter values:

557.1	26.13	79.7	145.3	0.01368
-------	-------	------	-------	---------

Iteration = 2

Total time = 6.9231s Reduced Chi^2 = 1.1175 Levenberg-Marquardt = 0.01

Free parameter values:

1032	32.48	114.7	189.4	0.01186
------	-------	-------	-------	---------

Total time = 7.5317s Reduced Chi^2 = 1.0534 Levenberg-Marquardt = 0.1

Free parameter values:

714.8	28.38	91.02	158.7	0.01227
-------	-------	-------	-------	---------

Iteration = 3

```

-----
Total time = 10.6695s    Reduced Chi^2 = 1.1141    Levenberg-Marquardt = 0.01
Free parameter values:
    1351          35.45          123.8          212.6          0.01179
Total time = 11.329s    Reduced Chi^2 = 1.0555    Levenberg-Marquardt = 0.1
Free parameter values:
    924.9          30.8          105.7          175.2          0.01201
Total time = 11.9941s    Reduced Chi^2 = 1.0547    Levenberg-Marquardt = 1
Free parameter values:
    783.6          29.23          94.39          163.7          0.01217
Total time = 12.6436s    Reduced Chi^2 = 1.0553    Levenberg-Marquardt = 100
Free parameter values:
    722           28.48          91.13          159.2          0.01226
Total time = 13.2921s    Reduced Chi^2 = 1.0563    Levenberg-Marquardt = 10000
Free parameter values:
    715.5          28.39          91.03          158.7          0.01227
Total time = 13.9639s    Reduced Chi^2 = 1.0555    Levenberg-Marquardt = 1000000
Free parameter values:
    714.8          28.38          91.02          158.7          0.01227
*** No improvement in chi-squared over previous iteration ***
Total time = 13.9711s    Reduced Chi^2 = 1.0534
Free parameter values:
    714.8          28.38          91.02          158.7          0.01227
-----

```

Fit converged:

Parameter values (free parameters with error estimates):

Foreground parameter values:

```

-----
1          1
2          8
3          714.8 +/- 5118
4          28.38 +/- 82.66
5          91.02 +/- 572.7
6          158.7 +/- 286.7
7          0
8          0
9          0
10         0

```

Background parameter values:

```

-----
1          0.01227 +/- 0.001293
2          0
3          0

```

Covariance matrix for free parameters:

```

-----
1.0000 -0.9580 -0.9780 0.9743 0.2164
-0.9580 1.0000 0.8772 -0.8690 -0.0232
-0.9780 0.8772 1.0000 -0.9998 -0.3577
0.9743 -0.8690 -0.9998 1.0000 0.3672
0.2164 -0.0232 -0.3577 0.3672 1.0000

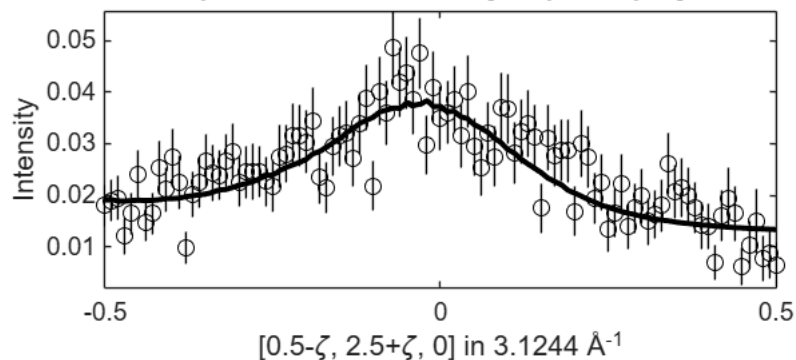
```

Fe_ei800_align_sym3.sqw

i= 28± 83; J0=1.6e+02±2.9e+02; J1= 0± 0; gamma=714.757±5117

$\xi \leq 0.1$ in $[0.5+\xi, 2.5+\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[0.5, 2.5, -\eta]$, $270 \leq$

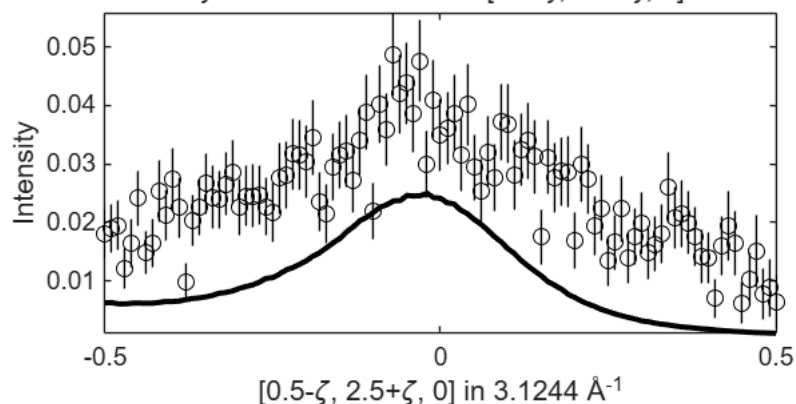
$\zeta=-0.505:0.01:0.505$ in $[0.5-\zeta, 2.5+\zeta, 0]$



Fe_ei800_align_sym3.sqw

$\xi \leq 0.1$ in $[0.5+\xi, 2.5+\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[0.5, 2.5, -\eta]$, $270 \leq$

$\zeta=-0.505:0.01:0.505$ in $[0.5-\zeta, 2.5+\zeta, 0]$



**** En = 285±5 going to 405

Beginning fit (max 20 iterations)-----

Starting point

Total time = 0.70524s Reduced Chi^2 = 10.2155

Free parameter values:

714.8	28.38	91.02	158.7	0.02377
-------	-------	-------	-------	---------

Iteration = 1

Total time = 4.1878s Reduced Chi^2 = 1.3804 Levenberg-Marquardt = 0.1

Free parameter values:

997.3	25.52	170.4	172.7	0.01181
-------	-------	-------	-------	---------

Iteration = 2

Total time = 7.5346s Reduced Chi^2 = 2.7659 Levenberg-Marquardt = 0.01

Free parameter values:

2684	16.09	361.3	212.7	0.009247
------	-------	-------	-------	----------

Total time = 8.1774s Reduced Chi^2 = 1.3865 Levenberg-Marquardt = 0.1

Free parameter values:

1587	23.17	255.5	182.5	0.01016
------	-------	-------	-------	---------

Total time = 8.8399s Reduced Chi^2 = 1.351 Levenberg-Marquardt = 1

Free parameter values:

1191	24.95	192.9	175.4	0.01103
------	-------	-------	-------	---------

Iteration = 3

Total time = 12.1984s	Reduced Chi^2 = 1.3639		Levenberg-Marquardt = 0.1	
Free parameter values:				
1823	22.4	260.2	185.9	0.00991
Total time = 12.852s	Reduced Chi^2 = 1.3354		Levenberg-Marquardt = 1	
Free parameter values:				
1398	24.27	211.9	178.3	0.01054

Iteration = 4

Total time = 16.1872s	Reduced Chi^2 = 1.3702		Levenberg-Marquardt = 0.1	
Free parameter values:				
2155	20.99	253.9	191.9	0.009709
Total time = 16.8938s	Reduced Chi^2 = 1.3257		Levenberg-Marquardt = 1	
Free parameter values:				
1645	23.32	224	182.3	0.0102

Iteration = 5

Total time = 20.2641s	Reduced Chi^2 = 1.3606		Levenberg-Marquardt = 0.1	
Free parameter values:				
2477	19.87	241.9	196.2	0.009548
Total time = 20.9269s	Reduced Chi^2 = 1.3147		Levenberg-Marquardt = 1	
Free parameter values:				
1917	22.29	229.1	186.4	0.009945

Iteration = 6

Total time = 24.1114s	Reduced Chi^2 = 1.3453		Levenberg-Marquardt = 0.1	
Free parameter values:				
2795	18.75	219.3	199.6	0.009391
Total time = 24.7802s	Reduced Chi^2 = 1.3088		Levenberg-Marquardt = 1	
Free parameter values:				
2205	21.22	225.5	190.3	0.009737

Iteration = 7

Total time = 28.1401s	Reduced Chi^2 = 1.338		Levenberg-Marquardt = 0.1	
Free parameter values:				
3200	17.93	183.5	206.6	0.009253
Total time = 28.828s	Reduced Chi^2 = 1.3036		Levenberg-Marquardt = 1	
Free parameter values:				
2531	20.22	211.8	195.2	0.009557

Iteration = 8

Total time = 32.1705s	Reduced Chi^2 = 1.3246		Levenberg-Marquardt = 0.1	
Free parameter values:				
3577	17.01	133.5	212	0.009131
Total time = 32.8436s	Reduced Chi^2 = 1.2949		Levenberg-Marquardt = 1	
Free parameter values:				
2876	19.23	186.7	200.2	0.009399

Iteration = 9

Total time = 36.2527s	Reduced Chi^2 = 1.3103		Levenberg-Marquardt = 0.1	
Free parameter values:				
4092	16.88	78.39	222.8	0.008992
Total time = 36.9358s	Reduced Chi^2 = 1.2876		Levenberg-Marquardt = 1	
Free parameter values:				
3279	18.51	152	207.1	0.009245

Iteration = 10

Total time = 40.3359s	Reduced Chi^2 = 1.2924	Levenberg-Marquardt = 0.1
Free parameter values:		
4474	16.66	13.43
230.5	0.008896	
Total time = 41.0263s	Reduced Chi^2 = 1.2784	Levenberg-Marquardt = 1
Free parameter values:		
3678	17.93	107.4
214.2	0.009109	

Iteration = 11

Total time = 44.4924s	Reduced Chi^2 = 1.2777	Levenberg-Marquardt = 0.1
Free parameter values:		
5015	16.74	-66.61
243.5	0.008795	

Fit converged:

Parameter values (free parameters with error estimates):

Foreground parameter values:

1	1		
2	8		
3	5015	+/-	1.732e+06
4	16.74	+/-	3036
5	-66.61	+/-	2.083e+04
6	243.5	+/-	4.305e+04
7	0		
8	0		
9	0		
10	0		

Background parameter values:

1	0.008795	+/-	0.0015
2	0		
3	0		

Covariance matrix for free parameters:

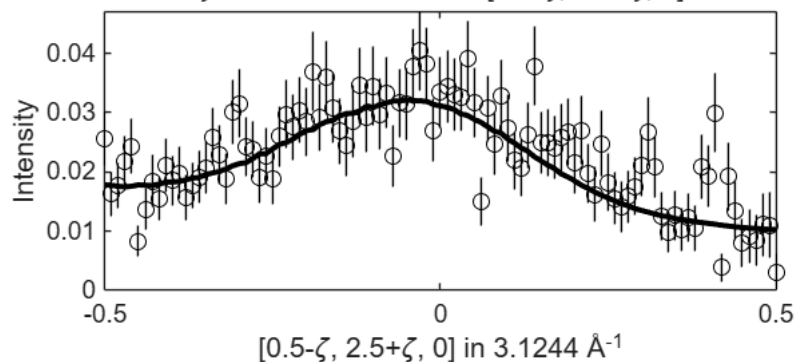
1.0000	1.0000	-0.9992	1.0000	-0.5119
1.0000	1.0000	-0.9991	1.0000	-0.5123
-0.9992	-0.9991	1.0000	-0.9992	0.5042
1.0000	1.0000	-0.9992	1.0000	-0.5116
-0.5119	-0.5123	0.5042	-0.5116	1.0000

Fe_ei800_align_sym3.sqw

$7 \pm 3e+03$; $J_0 = 2.4e+02 \pm 4.3e+04$; $J_1 = 0 \pm 0$; $\gamma = 5014.81 \pm 1.732$.

$\xi \leq 0.1$ in $[0.5 + \xi, 2.5 + \xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[0.5, 2.5, -\eta]$, $280 \leq$

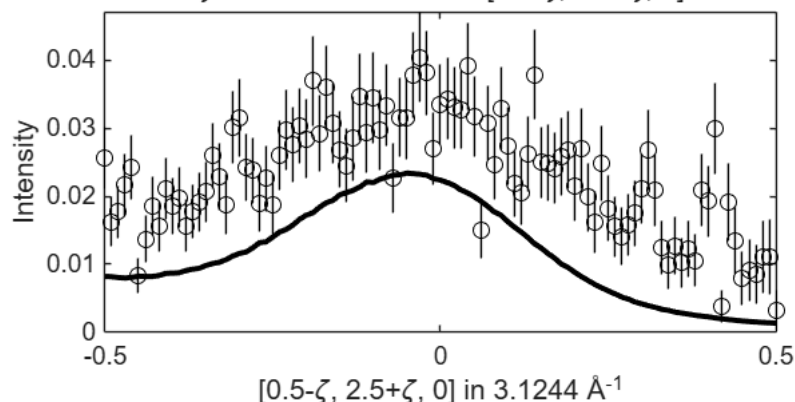
$\zeta = -0.505:0.01:0.505$ in $[0.5 - \zeta, 2.5 + \zeta, 0]$



Fe_ei800_align_sym3.sqw

$\xi \leq 0.1$ in $[0.5 + \xi, 2.5 + \xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[0.5, 2.5, -\eta]$, $280 \leq$

$\zeta = -0.505:0.01:0.505$ in $[0.5 - \zeta, 2.5 + \zeta, 0]$



**** En = 295±5 going to 405

Beginning fit (max 20 iterations)-----

Starting point

Total time = 0.69717s Reduced Chi^2 = 12.3539

Free parameter values:

5015 16.74 -66.61 243.5 0.02291

Iteration = 1

Total time = 4.2217s Reduced Chi^2 = 1.221 Levenberg-Marquardt = 0.1

Free parameter values:

5129 18.44 -179.7 281 0.01017

Iteration = 2

Total time = 7.6658s Reduced Chi^2 = 1.2135 Levenberg-Marquardt = 0.01

Free parameter values:

9534 23.67 -403 370.9 0.007515

Iteration = 3

Total time = 11.037s Reduced Chi^2 = 1.5161 Levenberg-Marquardt = 0.001

Free parameter values:

3.209e+04	49.97	-1174	803.3	0.007466
Total time = 11.7522s	Reduced Chi^2 = 1.1717		Levenberg-Marquardt = 0.01	
Free parameter values:				
1.713e+04	31.77	-729.3	505.6	0.007446

Iteration = 4

Total time = 15.2583s	Reduced Chi^2 = 1.4743		Levenberg-Marquardt = 0.001	
Free parameter values:				
5.828e+04	68.85	-1711	1097	0.007466
Total time = 16.0188s	Reduced Chi^2 = 1.1624		Levenberg-Marquardt = 0.01	
Free parameter values:				
3.099e+04	43.38	-1075	689.9	0.007464

Iteration = 5

Total time = 19.6679s	Reduced Chi^2 = 1.4583		Levenberg-Marquardt = 0.001	
Free parameter values:				
1.039e+05	92.41	-2330	1466	0.007454
Total time = 20.3942s	Reduced Chi^2 = 1.1628		Levenberg-Marquardt = 0.01	
Free parameter values:				
5.604e+04	58.65	-1480	931.8	0.007461
Total time = 21.073s	Reduced Chi^2 = 1.1384		Levenberg-Marquardt = 0.1	
Free parameter values:				
4.059e+04	48.07	-1212	763	0.007498

Iteration = 6

Total time = 24.5499s	Reduced Chi^2 = 1.1589		Levenberg-Marquardt = 0.01	
Free parameter values:				
7.61e+04	68.23	-1773	1086	0.007456
Total time = 25.2376s	Reduced Chi^2 = 1.1384		Levenberg-Marquardt = 0.1	
Free parameter values:				
5.251e+04	54.38	-1397	864	0.00747
Total time = 25.9204s	Reduced Chi^2 = 1.1344		Levenberg-Marquardt = 1	
Free parameter values:				
4.471e+04	50.09	-1276	794.1	0.007504

Iteration = 7

Total time = 29.3711s	Reduced Chi^2 = 1.137		Levenberg-Marquardt = 0.1	
Free parameter values:				
5.798e+04	57.12	-1480	907.2	0.007452
Total time = 30.052s	Reduced Chi^2 = 1.1351		Levenberg-Marquardt = 1	
Free parameter values:				
4.906e+04	52.31	-1343	829.6	0.007484
Total time = 30.7312s	Reduced Chi^2 = 1.1347		Levenberg-Marquardt = 10	
Free parameter values:				
4.612e+04	50.8	-1298	805.2	0.0075
Total time = 31.4004s	Reduced Chi^2 = 1.1325		Levenberg-Marquardt = 1000	
Free parameter values:				
4.485e+04	50.16	-1279	795.3	0.007503

Iteration = 8

Total time = 34.9583s	Reduced Chi^2 = 1.1351		Levenberg-Marquardt = 100	
Free parameter values:				
4.523e+04	50.35	-1285	798.3	0.007502
Total time = 35.6818s	Reduced Chi^2 = 1.1331		Levenberg-Marquardt = 1000	
Free parameter values:				
4.497e+04	50.22	-1281	796.2	0.007503
Total time = 36.3639s	Reduced Chi^2 = 1.1378		Levenberg-Marquardt = 10000	
Free parameter values:				
4.489e+04	50.18	-1279	795.6	0.007503

```

Total time = 37.1006s    Reduced Chi^2 = 1.14    Levenberg-Marquardt = 1000000
Free parameter values:
  4.485e+04    50.16    -1279    795.3    0.007503
Total time = 37.7953s    Reduced Chi^2 = 1.1366    Levenberg-Marquardt = 100000000
Free parameter values:
  4.485e+04    50.16    -1279    795.3    0.007503
Total time = 38.5631s    Reduced Chi^2 = 1.1339    Levenberg-Marquardt = 10000000000
Free parameter values:
  4.485e+04    50.16    -1279    795.3    0.007503
*** No improvement in chi-squared over previous iteration ***
Total time = 38.5714s    Reduced Chi^2 = 1.1325
Free parameter values:
  4.485e+04    50.16    -1279    795.3    0.007503

```

Fit converged:

Parameter values (free parameters with error estimates):

Foreground parameter values:

1	1		
2	8		
3	4.485e+04	+/-	2.449e+04
4	50.16	+/-	24.34
5	-1279	+/-	1441
6	795.3	+/-	121.9
7	0		
8	0		
9	0		
10	0		

Background parameter values:

1	0.007503	+/-	0.001126
2	0		
3	0		

Covariance matrix for free parameters:

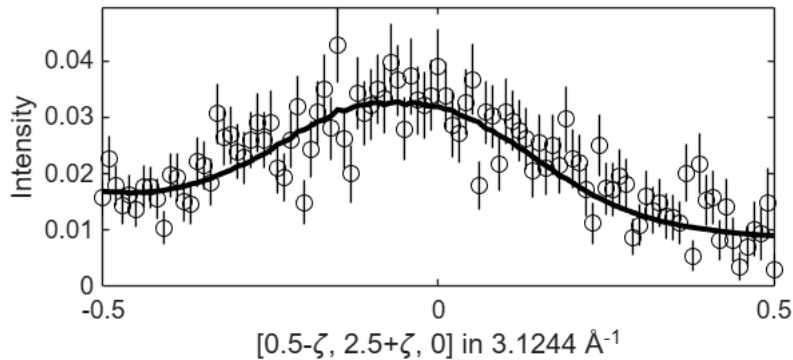
1.0000	0.9675	0.9781	0.9642	-0.4056
0.9675	1.0000	0.9872	0.9918	-0.3520
0.9781	0.9872	1.0000	0.9923	-0.2534
0.9642	0.9918	0.9923	1.0000	-0.2833
-0.4056	-0.3520	-0.2534	-0.2833	1.0000

Fe_ei800_align_sym3.sqw

S= 50± 24; J0=8e+02±1.2e+02; J1= 0± 0; gamma=44848.4±24485

$\xi \leq 0.1$ in $[0.5+\xi, 2.5+\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[0.5, 2.5, -\eta]$, $290 \leq$

$\zeta = -0.505:0.01:0.505$ in $[0.5-\zeta, 2.5+\zeta, 0]$



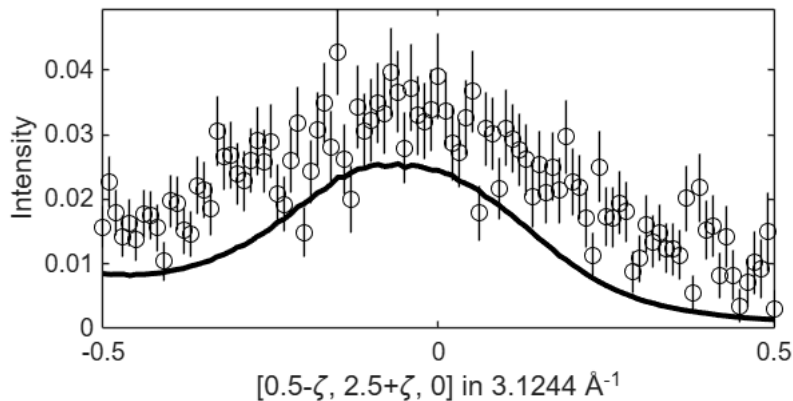
Warning: Connection to graphics acceleration hardware was lost. Graphics quality and performance might be diminished. See MATLAB System Requirements.

Warning: Unable to generate graphics because of system configuration or graphics resource constraint. See MATLAB System Requirements or contact MathWorks technical support.

Fe_ei800_align_sym3.sqw

$\xi \leq 0.1$ in $[0.5+\xi, 2.5+\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[0.5, 2.5, -\eta]$, $290 \leq$

$\zeta = -0.505:0.01:0.505$ in $[0.5-\zeta, 2.5+\zeta, 0]$



**** En = 305±5 going to 405

Beginning fit (max 20 iterations)-----

Starting point

Total time = 0.80033s Reduced Chi^2 = 14.5501

Free parameter values:

4.485e+04	50.16	-1279	795.3	0.02231
-----------	-------	-------	-------	---------

Iteration = 1

Total time = 4.6301s Reduced Chi^2 = 1.2894 Levenberg-Marquardt = 0.1

Free parameter values:

4.512e+04	37.74	-2404	899.3	0.01069
-----------	-------	-------	-------	---------

Iteration = 2

Total time = 8.4822s Reduced Chi^2 = 1.5355 Levenberg-Marquardt = 0.01

Free parameter values:

5.672e+04	34.5	-6387	1172	0.008543
-----------	------	-------	------	----------

Total time = 9.2975s Reduced Chi^2 = 1.2101 Levenberg-Marquardt = 0.1

```

Free parameter values:
  4.869e+04      36.3      -4183      977.1      0.008665
-----
Iteration = 3
-----
Total time = 13.1718s      Reduced Chi^2 = 1.1373      Levenberg-Marquardt = 0.01
Free parameter values:
  3.201e+04      35.34      -5544      1021      0.008739
-----
Iteration = 4
-----
Total time = 16.9679s      Reduced Chi^2 = 1.1893      Levenberg-Marquardt = 0.001
Free parameter values:
  6.779e+04      54.09      -8281      1547      0.008749
Total time = 17.7243s      Reduced Chi^2 = 1.1333      Levenberg-Marquardt = 0.01
Free parameter values:
  4.495e+04      41.38      -6328      1185      0.008769
-----
Iteration = 5
-----
Total time = 21.5854s      Reduced Chi^2 = 1.13      Levenberg-Marquardt = 0.001
Free parameter values:
  4.65e+04      41.26      -6264      1178      0.008759
-----
Iteration = 6
-----
Total time = 25.5443s      Reduced Chi^2 = 1.9051      Levenberg-Marquardt = 0.0001
Free parameter values:
  9266      24.64      -3739      705.1      0.008761
Total time = 26.3007s      Reduced Chi^2 = 1.1282      Levenberg-Marquardt = 0.001
Free parameter values:
  3.47e+04      36      -5464      1029      0.008761
-----
Iteration = 7
-----
Total time = 30.1944s      Reduced Chi^2 = 1.2061      Levenberg-Marquardt = 0.0001
Free parameter values:
  7.917e+04      58.93      -8951      1680      0.00874
Total time = 30.984s      Reduced Chi^2 = 1.1331      Levenberg-Marquardt = 0.001
Free parameter values:
  4.916e+04      43.22      -6568      1234      0.008741
Total time = 31.7657s      Reduced Chi^2 = 1.1258      Levenberg-Marquardt = 0.01
Free parameter values:
  3.964e+04      38.24      -5817      1092      0.008743
-----
Iteration = 8
-----
Total time = 35.6532s      Reduced Chi^2 = 1.1943      Levenberg-Marquardt = 0.001
Free parameter values:
  8.777e+04      61.54      -9340      1753      0.008739
Total time = 36.4262s      Reduced Chi^2 = 1.1272      Levenberg-Marquardt = 0.01
Free parameter values:
  5.505e+04      45.63      -6924      1301      0.008741
Total time = 37.196s      Reduced Chi^2 = 1.1282      Levenberg-Marquardt = 0.1
Free parameter values:
  4.463e+04      40.57      -6166      1158      0.008746
Total time = 37.9739s      Reduced Chi^2 = 1.1282      Levenberg-Marquardt = 10
Free parameter values:
  4.017e+04      38.47      -5854      1099      0.008745
Total time = 38.7445s      Reduced Chi^2 = 1.1282      Levenberg-Marquardt = 1000
Free parameter values:
  3.97e+04      38.27      -5821      1093      0.008743
Total time = 39.5745s      Reduced Chi^2 = 1.1311      Levenberg-Marquardt = 100000
Free parameter values:

```

```

3.965e+04      38.24      -5817      1092      0.008743
*** No improvement in chi-squared over previous iteration ***
Total time = 39.5815s      Reduced Chi^2 = 1.1258
Free parameter values:
3.964e+04      38.24      -5817      1092      0.008743

```

Fit converged:

Parameter values (free parameters with error estimates):

Foreground parameter values:

1	1		
2	8		
3	3.964e+04	+/-	7519
4	38.24	+/-	4.13
5	-5817	+/-	926.9
6	1092	+/-	42.64
7	0		
8	0		
9	0		
10	0		

Background parameter values:

1	0.008743	+/-	0.001127
2	0		
3	0		

Covariance matrix for free parameters:

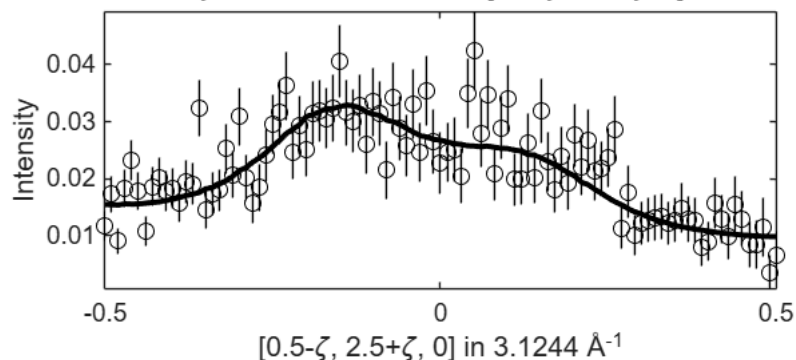
1.0000	0.7569	0.9603	-0.9262	-0.6383
0.7569	1.0000	0.6796	-0.6594	-0.9126
0.9603	0.6796	1.0000	-0.9433	-0.4606
-0.9262	-0.6594	-0.9433	1.0000	0.4465
-0.6383	-0.9126	-0.4606	0.4465	1.0000

Fe_ei800_align_sym3.sqw

S= 38± 4.1; J0=1.1e+03± 43; J1= 0± 0; gamma=39643.1±7518.9

$\xi \leq 0.1$ in $[0.5+\xi, 2.5+\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[0.5, 2.5, -\eta]$, $300 \leq$

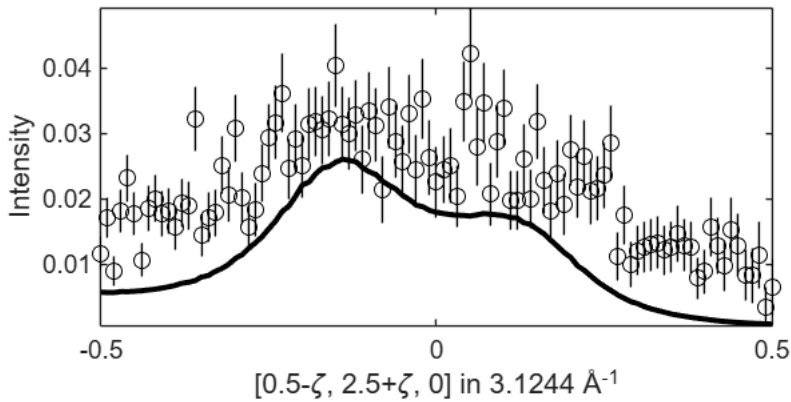
$\zeta = -0.505:0.01:0.505$ in $[0.5-\zeta, 2.5+\zeta, 0]$



Fe_ei800_align_sym3.sqw

$\xi \leq 0.1$ in $[0.5+\xi, 2.5+\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[0.5, 2.5, -\eta]$, $300 \leq$

$\zeta = -0.505:0.01:0.505$ in $[0.5-\zeta, 2.5+\zeta, 0]$



**** En = 315±5 going to 405

Beginning fit (max 20 iterations)-----

Starting point

Total time = 0.83815s Reduced Chi^2 = 12.393

Free parameter values:

3.964e+04	38.24	-5817	1092	0.0213
-----------	-------	-------	------	--------

Iteration = 1

Total time = 5.0386s Reduced Chi^2 = 1.0712 Levenberg-Marquardt = 0.1

Free parameter values:

3.813e+04	32.69	-5185	1060	0.01027
-----------	-------	-------	------	---------

Iteration = 2

Total time = 9.1106s Reduced Chi^2 = 1.0061 Levenberg-Marquardt = 0.01

Free parameter values:

4.18e+04	37.98	-4110	962.4	0.008015
----------	-------	-------	-------	----------

Iteration = 3

Total time = 13.1129s Reduced Chi^2 = 1.007 Levenberg-Marquardt = 0.001

Free parameter values:

4.612e+04	40.6	-4111	994.8	0.007708
-----------	------	-------	-------	----------

Total time = 13.9775s Reduced Chi^2 = 1.0027 Levenberg-Marquardt = 0.01

Free parameter values:

4.368e+04	39.45	-4031	970.4	0.007733
-----------	-------	-------	-------	----------

Iteration = 4

Total time = 18.1241s Reduced Chi^2 = 1.0033 Levenberg-Marquardt = 0.001

Free parameter values:

4.741e+04	41.22	-4151	1007	0.007705
-----------	-------	-------	------	----------

Total time = 18.9613s Reduced Chi^2 = 1.005 Levenberg-Marquardt = 0.01

Free parameter values:

4.503e+04	40.14	-4050	981.6	0.007709
-----------	-------	-------	-------	----------

Total time = 19.7954s Reduced Chi^2 = 1.0054 Levenberg-Marquardt = 0.1

Free parameter values:

4.414e+04	39.71	-4032	973.8	0.007721
-----------	-------	-------	-------	----------

Total time = 20.6298s Reduced Chi^2 = 1.004 Levenberg-Marquardt = 10

Free parameter values:

```

4.372e+04      39.49      -4031      970.7      0.007732
Total time = 21.4739s      Reduced Chi^2 = 1.004      Levenberg-Marquardt = 1000
Free parameter values:
4.368e+04      39.46      -4031      970.4      0.007733
Total time = 22.2993s      Reduced Chi^2 = 1.0056      Levenberg-Marquardt = 100000
Free parameter values:
4.368e+04      39.45      -4031      970.4      0.007733
*** No improvement in chi-squared over previous iteration ***
Total time = 22.3097s      Reduced Chi^2 = 1.0027
Free parameter values:
4.368e+04      39.45      -4031      970.4      0.007733

```

Fit converged:

Parameter values (free parameters with error estimates):

Foreground parameter values:

```

-----
1          1
2          8
3      4.368e+04 +/- 8687
4          39.45 +/- 5.027
5          -4031 +/- 906
6          970.4 +/- 23.21
7          0
8          0
9          0
10         0

```

Background parameter values:

```

-----
1      0.007733 +/- 0.001014
2          0
3          0

```

Covariance matrix for free parameters:

```

-----
1.0000    0.8702    0.9501   -0.8211   -0.5840
0.8702    1.0000    0.8100   -0.6109   -0.7917
0.9501    0.8100    1.0000   -0.8438   -0.3600
-0.8211   -0.6109   -0.8438    1.0000    0.2100
-0.5840   -0.7917   -0.3600    0.2100    1.0000

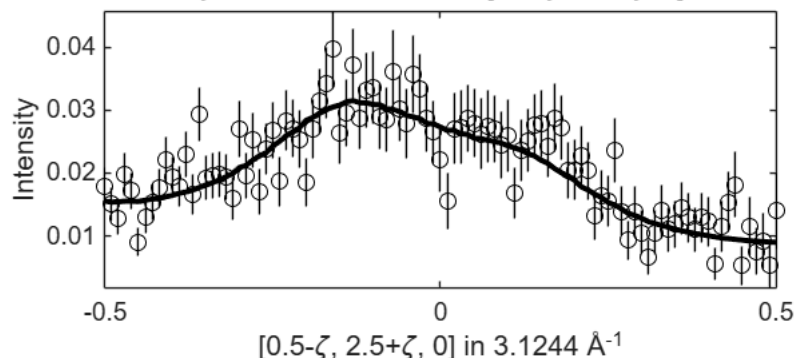
```

Fe_ei800_align_sym3.sqw

S= 39± 5; J0=9.7e+02± 23; J1= 0± 0; gamma=43679.2±8687.4

$\xi \leq 0.1$ in $[0.5+\xi, 2.5+\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[0.5, 2.5, -\eta]$, $310 \leq$

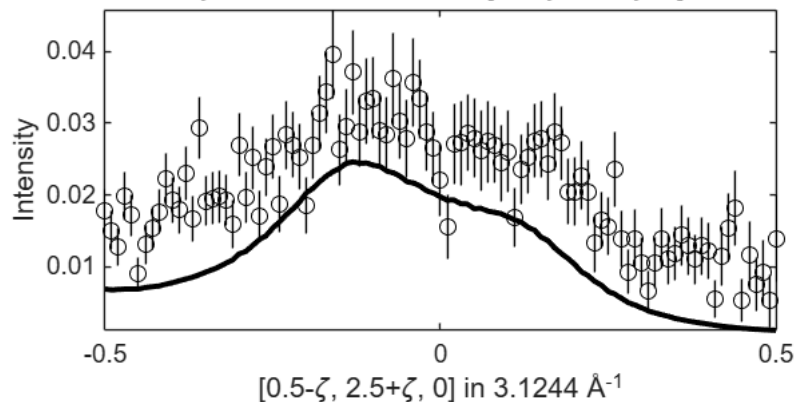
$\zeta = -0.505:0.01:0.505$ in $[0.5-\zeta, 2.5+\zeta, 0]$



Fe_ei800_align_sym3.sqw

$\xi \leq 0.1$ in $[0.5+\xi, 2.5+\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[0.5, 2.5, -\eta]$, $310 \leq$

$\zeta = -0.505:0.01:0.505$ in $[0.5-\zeta, 2.5+\zeta, 0]$



**** En = 325±5 going to 405

Beginning fit (max 20 iterations)-----

Starting point

Total time = 0.89624s Reduced Chi^2 = 15.6642

Free parameter values:

4.368e+04	39.45	-4031	970.4	0.02012
-----------	-------	-------	-------	---------

Iteration = 1

Total time = 5.6204s Reduced Chi^2 = 1.2016

Levenberg-Marquardt = 0.1

Free parameter values:

5.031e+04	35.34	-3912	983.7	0.008388
-----------	-------	-------	-------	----------

Iteration = 2

Total time = 10.2404s Reduced Chi^2 = 1.119

Levenberg-Marquardt = 0.01

Free parameter values:

4.73e+04	37.97	-3162	855.4	0.005914
----------	-------	-------	-------	----------

Iteration = 3

Total time = 14.9737s Reduced Chi^2 = 5.4798

Levenberg-Marquardt = 0.001

Free parameter values:

-1.981e+04	11.15	-1075	269.9	0.005809
------------	-------	-------	-------	----------

Total time = 15.894s Reduced Chi^2 = 1.1581

Levenberg-Marquardt = 0.01

Free parameter values:

2.49e+04	29.22	-2556	673.2	0.005801
----------	-------	-------	-------	----------

Total time = 16.8593s Reduced Chi^2 = 1.1066

Levenberg-Marquardt = 0.1

Free parameter values:

3.956e+04	35.1	-2974	799.7	0.005795
-----------	------	-------	-------	----------

Iteration = 4

Total time = 21.5296s Reduced Chi^2 = 1.1164

Levenberg-Marquardt = 0.01

Free parameter values:

2.577e+04	28.85	-2504	661.2	0.005784
-----------	-------	-------	-------	----------

Total time = 22.4548s Reduced Chi^2 = 1.1099

Levenberg-Marquardt = 0.1

Free parameter values:

3.527e+04	33.09	-2840	755.5	0.00578
-----------	-------	-------	-------	---------

Total time = 23.4272s Reduced Chi^2 = 1.1088

Levenberg-Marquardt = 1

Free parameter values:

```

      3.83e+04      34.46      -2937      785.4      0.005788
Total time = 24.4351s      Reduced Chi^2 = 1.1101      Levenberg-Marquardt = 100
Free parameter values:
      3.945e+04      35.03      -2971      798.3      0.005796
Total time = 25.395s      Reduced Chi^2 = 1.1105      Levenberg-Marquardt = 10000
Free parameter values:
      3.955e+04      35.09      -2973      799.6      0.005795
Total time = 26.3202s      Reduced Chi^2 = 1.1107      Levenberg-Marquardt = 1000000
Free parameter values:
      3.956e+04      35.1      -2974      799.7      0.005795
*** No improvement in chi-squared over previous iteration ***
Total time = 26.3271s      Reduced Chi^2 = 1.1066
Free parameter values:
      3.956e+04      35.1      -2974      799.7      0.005795

```

Fit converged:

Parameter values (free parameters with error estimates):

Foreground parameter values:

```

-----
1          1
2          8
3      3.956e+04 +/- 8201
4          35.1 +/- 4.463
5          -2974 +/- 724.2
6          799.7 +/- 15.25
7           0
8           0
9           0
10          0

```

Background parameter values:

```

-----
1      0.005795 +/- 0.0009582
2           0
3           0

```

Covariance matrix for free parameters:

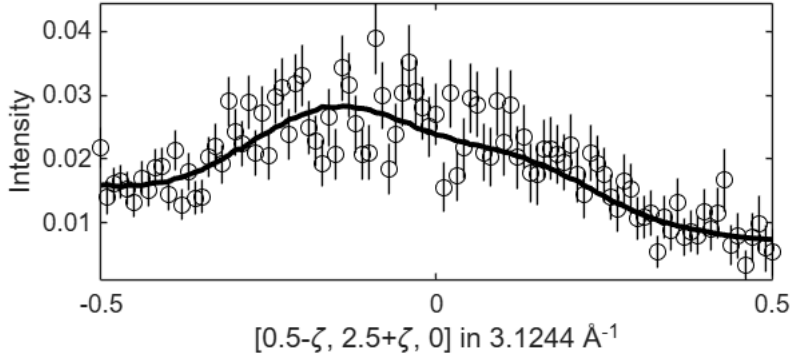
```

-----
1.0000  0.8677  0.9336 -0.1828 -0.5250
0.8677  1.0000  0.7412  0.1695 -0.7844
0.9336  0.7412  1.0000 -0.3852 -0.2231
-0.1828 0.1695 -0.3852  1.0000 -0.5835
-0.5250 -0.7844 -0.2231 -0.5835  1.0000

```

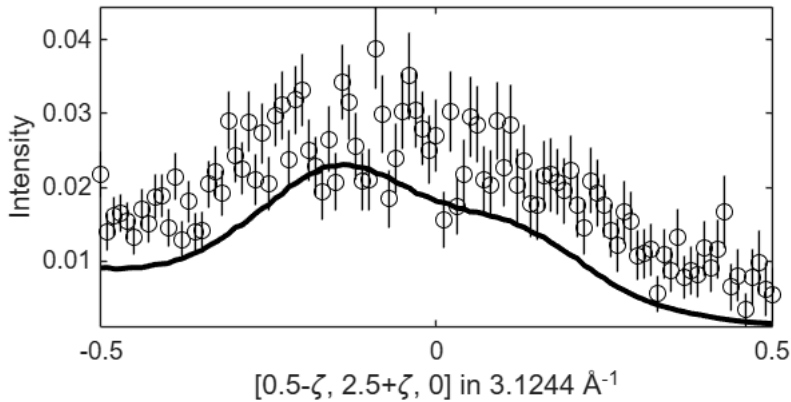
Fe_ei800_align_sym3.sqw

S= 35 ± 4.5 ; J0=8e+02 $\pm$ 15; J1= 0 $\pm$ 0; gamma=39560.7 $\pm$ 8200.74
 $\xi \leq 0.1$ in $[0.5+\xi, 2.5+\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[0.5, 2.5, -\eta]$, $320 \leq$
 $\zeta = -0.505:0.01:0.505$ in $[0.5-\zeta, 2.5+\zeta, 0]$



Fe_ei800_align_sym3.sqw

$\xi \leq 0.1$ in $[0.5+\xi, 2.5+\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[0.5, 2.5, -\eta]$, $320 \leq$
 $\zeta = -0.505:0.01:0.505$ in $[0.5-\zeta, 2.5+\zeta, 0]$



**** En = 335 $\pm$ 5 going to 405

Beginning fit (max 20 iterations)-----

Starting point

Total time = 1.4225s Reduced Chi^2 = 19.2087

Free parameter values:

3.956e+04	35.1	-2974	799.7	0.01985
-----------	------	-------	-------	---------

Iteration = 1

Total time = 8.3134s Reduced Chi^2 = 1.1913 Levenberg-Marquardt = 0.1

Free parameter values:

4.011e+04	29.91	-3692	864.5	0.007978
-----------	-------	-------	-------	----------

Iteration = 2

Total time = 15.1576s Reduced Chi^2 = 1.0896 Levenberg-Marquardt = 0.01

Free parameter values:

4.777e+04	37.09	-4880	999.2	0.005931
-----------	-------	-------	-------	----------

Iteration = 3

Total time = 21.9805s Reduced Chi^2 = 1.1022 Levenberg-Marquardt = 0.001

Free parameter values:

7.853e+04	49.83	-6426	1320	0.005875
Total time = 23.3393s	Reduced Chi^2 = 1.0867		Levenberg-Marquardt = 0.01	
Free parameter values:				
5.744e+04	41.57	-5350	1099	0.005884

Iteration = 4

Total time = 30.1919s	Reduced Chi^2 = 1.1169		Levenberg-Marquardt = 0.001	
Free parameter values:				
1.012e+05	57.43	-7363	1514	0.005879
Total time = 31.5852s	Reduced Chi^2 = 1.089		Levenberg-Marquardt = 0.01	
Free parameter values:				
7.161e+04	46.64	-5978	1230	0.005882
Total time = 32.9818s	Reduced Chi^2 = 1.0886		Levenberg-Marquardt = 0.1	
Free parameter values:				
6.21e+04	43.19	-5549	1140	0.005888
Total time = 34.3936s	Reduced Chi^2 = 1.0877		Levenberg-Marquardt = 10	
Free parameter values:				
5.796e+04	41.74	-5373	1103	0.005888
Total time = 35.845s	Reduced Chi^2 = 1.0853		Levenberg-Marquardt = 1000	
Free parameter values:				
5.749e+04	41.59	-5353	1100	0.005885

Iteration = 5

Total time = 42.687s	Reduced Chi^2 = 1.089		Levenberg-Marquardt = 100	
Free parameter values:				
5.764e+04	41.64	-5359	1101	0.005886
Total time = 44.1363s	Reduced Chi^2 = 1.0862		Levenberg-Marquardt = 1000	
Free parameter values:				
5.754e+04	41.6	-5355	1100	0.005885
Total time = 45.5573s	Reduced Chi^2 = 1.0896		Levenberg-Marquardt = 10000	
Free parameter values:				
5.75e+04	41.59	-5353	1100	0.005885
Total time = 46.9618s	Reduced Chi^2 = 1.0839		Levenberg-Marquardt = 1000000	
Free parameter values:				
5.749e+04	41.59	-5353	1100	0.005885

Iteration = 6

Total time = 53.8372s	Reduced Chi^2 = 1.0881		Levenberg-Marquardt = 100000	
Free parameter values:				
5.75e+04	41.59	-5353	1100	0.005885
Total time = 55.2577s	Reduced Chi^2 = 1.0882		Levenberg-Marquardt = 1000000	
Free parameter values:				
5.749e+04	41.59	-5353	1100	0.005885
Total time = 56.6195s	Reduced Chi^2 = 1.0856		Levenberg-Marquardt = 10000000	
Free parameter values:				
5.749e+04	41.59	-5353	1100	0.005885
Total time = 58.0193s	Reduced Chi^2 = 1.0859		Levenberg-Marquardt = 1000000000	
Free parameter values:				
5.749e+04	41.59	-5353	1100	0.005885
Total time = 59.4068s	Reduced Chi^2 = 1.0897		Levenberg-Marquardt = 100000000000	
Free parameter values:				
5.749e+04	41.59	-5353	1100	0.005885
Total time = 60.7547s	Reduced Chi^2 = 1.0885		Levenberg-Marquardt = 10000000000000	
Free parameter values:				
5.749e+04	41.59	-5353	1100	0.005885
*** No improvement in chi-squared over previous iteration ***				
Total time = 60.7621s	Reduced Chi^2 = 1.0839			
Free parameter values:				
5.749e+04	41.59	-5353	1100	0.005885

Fit converged:

Parameter values (free parameters with error estimates):

 Foreground parameter values:

1	1		
2	8		
3	5.749e+04	+/-	9335
4	41.59	+/-	4.053
5	-5353	+/-	722.9
6	1100	+/-	26.34
7	0		
8	0		
9	0		
10	0		

Background parameter values:

1	0.005885	+/-	0.0009123
2	0		
3	0		

Covariance matrix for free parameters:

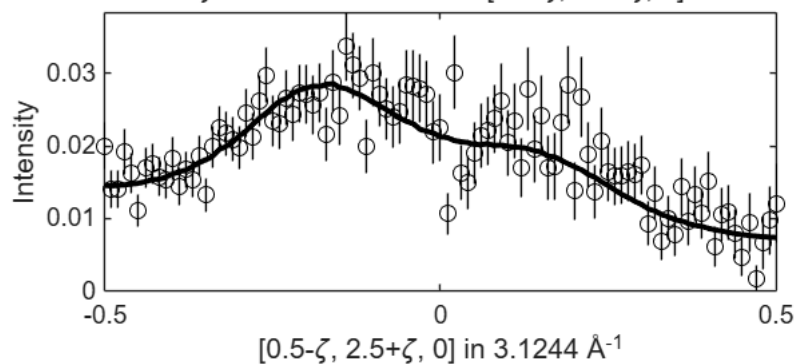
1.0000	0.7688	0.9157	-0.5831	-0.6422
0.7688	1.0000	0.5670	-0.0969	-0.9252
0.9157	0.5670	1.0000	-0.7876	-0.3221
-0.5831	-0.0969	-0.7876	1.0000	-0.1701
-0.6422	-0.9252	-0.3221	-0.1701	1.0000

Fe_ei800_align_sym3.sqw

S= 42± 4.1; J0=1.1e+03± 26; J1= 0± 0; gamma=57489.9±9334.6

$\xi \leq 0.1$ in $[0.5+\xi, 2.5+\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[0.5, 2.5, -\eta]$, $330 \leq$

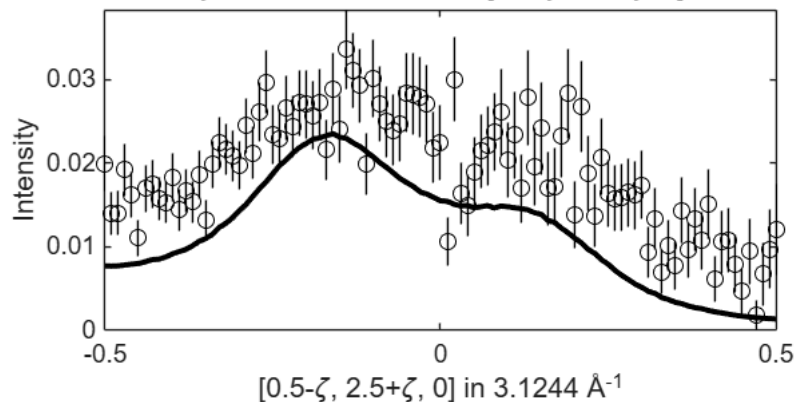
$\zeta = -0.505:0.01:0.505$ in $[0.5-\zeta, 2.5+\zeta, 0]$



Fe_ei800_align_sym3.sqw

$\xi \leq 0.1$ in $[0.5+\xi, 2.5+\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[0.5, 2.5, -\eta]$, $330 \leq$

$\zeta = -0.505:0.01:0.505$ in $[0.5-\zeta, 2.5+\zeta, 0]$



**** En = 345±5 going to 405

Beginning fit (max 20 iterations)-----

Starting point

Total time = 1.4748s Reduced Chi^2 = 18.2549

Free parameter values:

5.749e+04	41.59	-5353	1100	0.01984
-----------	-------	-------	------	---------

Iteration = 1

Total time = 8.7041s Reduced Chi^2 = 1.0228

Levenberg-Marquardt = 0.1

Free parameter values:

6.144e+04	32.07	-5121	1114	0.009509
-----------	-------	-------	------	----------

Iteration = 2

Total time = 16.0304s Reduced Chi^2 = 0.96589

Levenberg-Marquardt = 0.01

Free parameter values:

5.239e+04	33.01	-3791	929.5	0.007573
-----------	-------	-------	-------	----------

Iteration = 3

Total time = 23.2814s Reduced Chi^2 = 1.6045

Levenberg-Marquardt = 0.001

Free parameter values:

1.284e+04	20.42	-2309	574.8	0.007387
-----------	-------	-------	-------	----------

Total time = 24.7481s Reduced Chi^2 = 0.96134

Levenberg-Marquardt = 0.01

Free parameter values:

4.001e+04	29.02	-3309	817.4	0.007394
-----------	-------	-------	-------	----------

Iteration = 4

Total time = 31.9579s Reduced Chi^2 = 1.2545

Levenberg-Marquardt = 0.001

Free parameter values:

1.275e+04	18.81	-2133	529.9	0.007385
-----------	-------	-------	-------	----------

Total time = 33.4271s Reduced Chi^2 = 0.96172

Levenberg-Marquardt = 0.01

Free parameter values:

3.178e+04	25.77	-2934	725.5	0.007387
-----------	-------	-------	-------	----------

Total time = 34.922s Reduced Chi^2 = 0.963

Levenberg-Marquardt = 0.1

Free parameter values:

3.771e+04	27.95	-3195	787.6	0.00739
-----------	-------	-------	-------	---------

Total time = 36.3984s Reduced Chi^2 = 0.96183

Levenberg-Marquardt = 10

Free parameter values:

3.99e+04	28.9	-3303	814	0.007392
Total time = 37.825s	Reduced Chi^2 = 0.96276		Levenberg-Marquardt = 1000	
Free parameter values:				
4e+04	29.01	-3309	817	0.007394
Total time = 39.3094s	Reduced Chi^2 = 0.96106		Levenberg-Marquardt = 100000	
Free parameter values:				
4.001e+04	29.02	-3309	817.3	0.007394

Fit converged:

Parameter values (free parameters with error estimates):

Foreground parameter values:

1	1		
2	8		
3	4.001e+04	+/-	3.164e+07
4	29.02	+/-	1.157e+04
5	-3309	+/-	1.324e+06
6	817.3	+/-	3.245e+05
7	0		
8	0		
9	0		
10	0		

Background parameter values:

1	0.007394	+/-	0.0009563
2	0		
3	0		

Covariance matrix for free parameters:

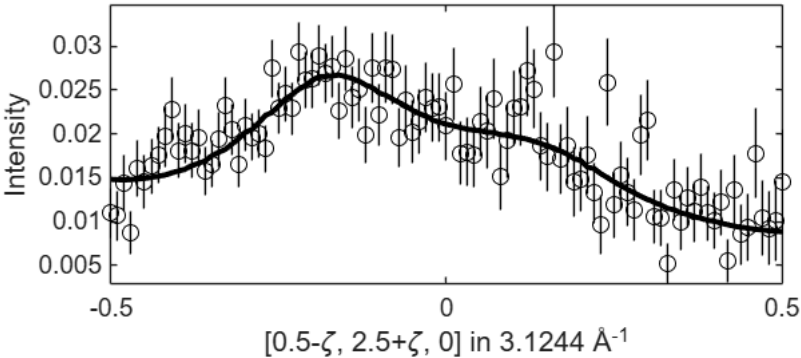
1.0000	1.0000	-1.0000	1.0000	-0.1783
1.0000	1.0000	-1.0000	1.0000	-0.1784
-1.0000	-1.0000	1.0000	-1.0000	0.1779
1.0000	1.0000	-1.0000	1.0000	-0.1781
-0.1783	-0.1784	0.1779	-0.1781	1.0000

Fe_ei800_align_sym3.sqw

!9±1.2e+04; J0=8.2e+02±3.2e+05; J1= 0± 0; gamma=40013±3.163i

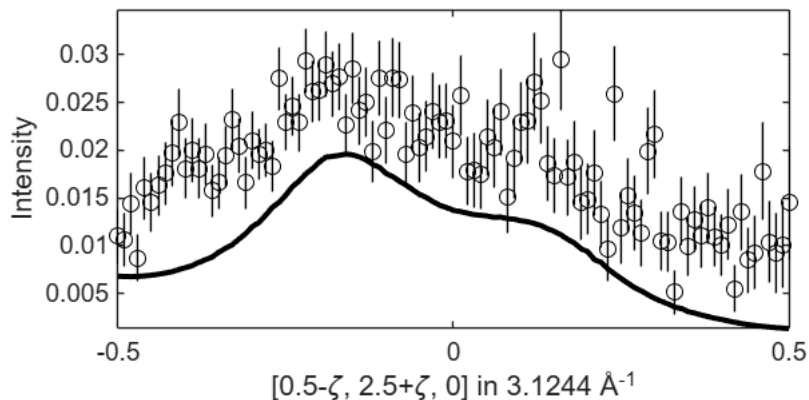
$\xi \leq 0.1$ in $[0.5+\xi, 2.5+\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[0.5, 2.5, -\eta]$, $340 \leq$

$\zeta=-0.505:0.01:0.505$ in $[0.5-\zeta, 2.5+\zeta, 0]$



Fe_ei800_align_sym3.sqw

$\xi \leq 0.1$ in $[0.5+\xi, 2.5+\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[0.5, 2.5, -\eta]$, $340 \leq \zeta = -0.505:0.01:0.505$ in $[0.5-\zeta, 2.5+\zeta, 0]$



 **** En = 355±5 going to 405

Beginning fit (max 20 iterations)-----
 Starting point

Total time = 1.4314s Reduced Chi^2 = 16.3489
 Free parameter values:
 4.001e+04 29.02 -3309 817.3 0.01962

Iteration = 1

Total time = 8.7245s Reduced Chi^2 = 1.1446 Levenberg-Marquardt = 0.1
 Free parameter values:
 3.406e+04 25.37 -3851 851.6 0.008436

Iteration = 2

Total time = 15.9667s Reduced Chi^2 = 1.0515 Levenberg-Marquardt = 0.01
 Free parameter values:
 3.239e+04 28.76 -3895 819.6 0.006303

Iteration = 3

Total time = 23.1813s Reduced Chi^2 = 1.0493 Levenberg-Marquardt = 0.001
 Free parameter values:
 3.421e+04 29.68 -3886 835.2 0.00617

Iteration = 4

Total time = 30.3078s Reduced Chi^2 = 1.0504 Levenberg-Marquardt = 0.0001
 Free parameter values:
 4.172e+04 32.96 -4328 928.1 0.006177
 Total time = 31.7658s Reduced Chi^2 = 1.0484 Levenberg-Marquardt = 0.001
 Free parameter values:
 3.655e+04 30.7 -4031 864.7 0.006177

Fit converged:

Parameter values (free parameters with error estimates):

Foreground parameter values:

1	1		
2	8		
3	3.655e+04	+/-	1.664e+07
4	30.7	+/-	7052
5	-4031	+/-	9.274e+05
6	864.7	+/-	1.977e+05
7	0		
8	0		
9	0		
10	0		

Background parameter values:

1	0.006177	+/-	0.0009323
2	0		
3	0		

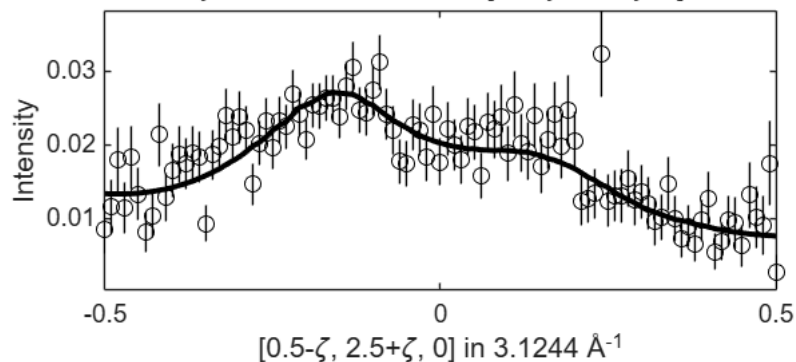
Covariance matrix for free parameters:

1.0000	1.0000	-1.0000	1.0000	-0.1518
1.0000	1.0000	-1.0000	1.0000	-0.1520
-1.0000	-1.0000	1.0000	-1.0000	0.1512
1.0000	1.0000	-1.0000	1.0000	-0.1515
-0.1518	-0.1520	0.1512	-0.1515	1.0000

Fe_ei800_align_sym3.sqw

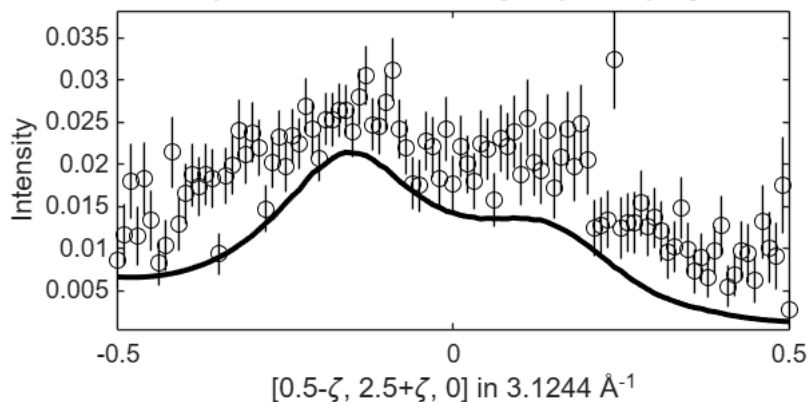
$\mu \pm 7.1e+03$; $J_0 = 8.6e+02 \pm 2e+05$; $J_1 = 0 \pm 0$; $\gamma = 36554.9 \pm 1.664$

$\xi \leq 0.1$ in $[0.5+\xi, 2.5+\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[0.5, 2.5, -\eta]$, $350 \leq \zeta = -0.505:0.01:0.505$ in $[0.5-\zeta, 2.5+\zeta, 0]$



Fe_ei800_align_sym3.sqw

$\xi \leq 0.1$ in $[0.5+\xi, 2.5+\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[0.5, 2.5, -\eta]$, $350 \leq \zeta = -0.505:0.01:0.505$ in $[0.5-\zeta, 2.5+\zeta, 0]$



**** En = 365±5 going to 405

Warning: Dataset:1 points with zero, negative or undefined (infinite or NaN) error bars have been removed from fit, which is 0.9901

Beginning fit (max 20 iterations)-----

Starting point

Total time = 0.9009s Reduced Chi^2 = 15.3417

Free parameter values:

3.655e+04	30.7	-4031	864.7	0.01906
-----------	------	-------	-------	---------

Iteration = 1

Total time = 5.7802s Reduced Chi^2 = 1.0241

Levenberg-Marquardt = 0.1

Free parameter values:

2.821e+04	22.74	-4808	880.8	0.008818
-----------	-------	-------	-------	----------

Iteration = 2

Total time = 10.5163s Reduced Chi^2 = 0.97555

Levenberg-Marquardt = 0.01

Free parameter values:

1.68e+04	22.84	-5072	822	0.007373
----------	-------	-------	-----	----------

Iteration = 3

Total time = 15.203s Reduced Chi^2 = 0.95068

Levenberg-Marquardt = 0.001

Free parameter values:

1.445e+04	19.68	-4206	705	0.007339
-----------	-------	-------	-----	----------

Iteration = 4

Total time = 20.0336s Reduced Chi^2 = 10.9894

Levenberg-Marquardt = 0.0001

Free parameter values:

-1.271e+04	0.8644	-256.8	44	0.007359
------------	--------	--------	----	----------

Total time = 20.9797s Reduced Chi^2 = 1.1694

Levenberg-Marquardt = 0.001

Free parameter values:

5694	13.68	-2973	496.1	0.007355
------	-------	-------	-------	----------

Total time = 21.9718s Reduced Chi^2 = 0.95087

Levenberg-Marquardt = 0.01

Free parameter values:

1.155e+04	17.76	-3828	639	0.007347
-----------	-------	-------	-----	----------

Total time = 22.9812s Reduced Chi^2 = 0.9475

Levenberg-Marquardt = 1

Free parameter values:

1.414e+04	19.51	-4170	698.4	0.007339
-----------	-------	-------	-------	----------

Iteration = 5

Total time = 27.8059s Reduced Chi^2 = 0.9518

Levenberg-Marquardt = 0.1

Free parameter values:

1.312e+04	18.86	-4046	675.9	0.007335
-----------	-------	-------	-------	----------

Total time = 28.8165s Reduced Chi^2 = 0.94861

Levenberg-Marquardt = 1

Free parameter values:

1.381e+04	19.31	-4132	691.2	0.007337
-----------	-------	-------	-------	----------

Total time = 29.8696s Reduced Chi^2 = 0.94759

Levenberg-Marquardt = 10

Free parameter values:

1.403e+04	19.45	-4158	696.1	0.007339
-----------	-------	-------	-------	----------

Total time = 30.9212s Reduced Chi^2 = 0.95033

Levenberg-Marquardt = 1000

Free parameter values:

1.413e+04	19.5	-4169	698.1	0.007339
-----------	------	-------	-------	----------

Total time = 31.9894s Reduced Chi^2 = 0.94896

Levenberg-Marquardt = 100000

Free parameter values:

1.414e+04	19.51	-4170	698.3	0.007339
-----------	-------	-------	-------	----------

Total time = 33.0603s Reduced Chi^2 = 0.94998

Levenberg-Marquardt = 10000000

```

Free parameter values:
  1.414e+04      19.51      -4170      698.4      0.007339
*** No improvement in chi-squared over previous iteration ***
Total time = 33.0689s      Reduced Chi^2 = 0.9475
Free parameter values:
  1.414e+04      19.51      -4170      698.4      0.007339

```

Fit converged:

Parameter values (free parameters with error estimates):

Foreground parameter values:

1	1		
2	8		
3	1.414e+04	+/-	2493
4	19.51	+/-	2.03
5	-4170	+/-	509.3
6	698.4	+/-	35.38
7	0		
8	0		
9	0		
10	0		

Background parameter values:

1	0.007339	+/-	0.0009305
2	0		
3	0		

Covariance matrix for free parameters:

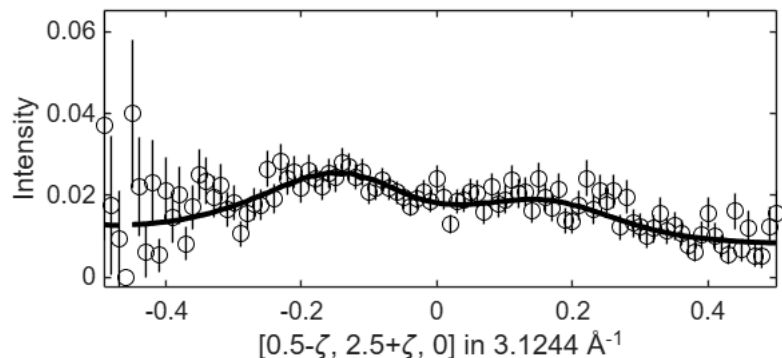
1.0000	0.7323	0.9789	-0.9356	-0.6760
0.7323	1.0000	0.6519	-0.6405	-0.9561
0.9789	0.6519	1.0000	-0.9671	-0.5476
-0.9356	-0.6405	-0.9671	1.0000	0.4964
-0.6760	-0.9561	-0.5476	0.4964	1.0000

Fe_ei800_align_sym3.sqw

S= 20± 2; J0=7e+02± 35; J1= 0± 0; gamma=14139.6±2493.31

$\xi \leq 0.1$ in $[0.5+\xi, 2.5+\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[0.5, 2.5, -\eta]$, $360 \leq$

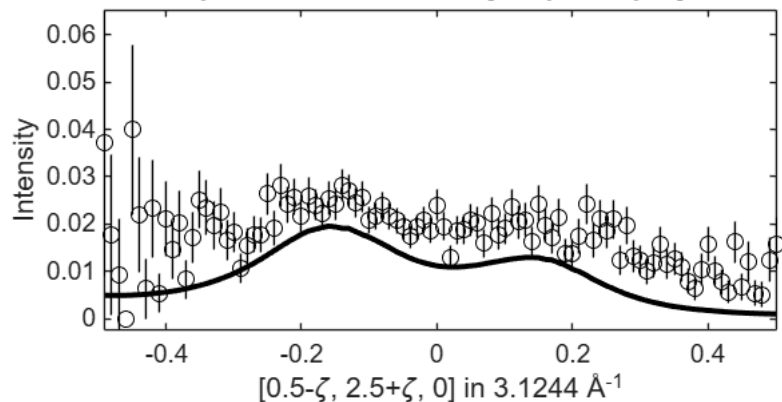
$\zeta = -0.505:0.01:0.505$ in $[0.5-\zeta, 2.5+\zeta, 0]$



Fe_ei800_align_sym3.sqw

$\xi \leq 0.1$ in $[0.5+\xi, 2.5+\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[0.5, 2.5, -\eta]$, $360 \leq$

$\zeta = -0.505:0.01:0.505$ in $[0.5-\zeta, 2.5+\zeta, 0]$



**** En = 375±5 going to 405

Warning: Dataset:2 points with zero, negative or undefined (infinite or NaN) error bars have been removed from fit, which is 1.9802

Beginning fit (max 20 iterations)-----

Starting point

Total time = 0.98799s Reduced Chi^2 = 13.1699

Free parameter values:

1.414e+04	19.51	-4170	698.4	0.0179
-----------	-------	-------	-------	--------

Iteration = 1

Total time = 5.8583s Reduced Chi^2 = 1.2495 Levenberg-Marquardt = 0.1

Free parameter values:

7257	12.34	-4199	657.1	0.008974
------	-------	-------	-------	----------

Iteration = 2

Total time = 10.7168s Reduced Chi^2 = 1.1899 Levenberg-Marquardt = 0.01

Free parameter values:

4303	11.76	-3556	528.6	0.007725
------	-------	-------	-------	----------

Iteration = 3

Total time = 15.8273s Reduced Chi^2 = 2.1353 Levenberg-Marquardt = 0.001

Free parameter values:

1081	6.402	-2140	323.7	0.007764
------	-------	-------	-------	----------

Total time = 16.7939s Reduced Chi^2 = 1.1815 Levenberg-Marquardt = 0.01

Free parameter values:

3345	9.731	-3103	464.9	0.007748
------	-------	-------	-------	----------

Iteration = 4

Total time = 21.7101s Reduced Chi^2 = 2.9172 Levenberg-Marquardt = 0.001

Free parameter values:

736.1	5.534	-1809	274.8	0.007774
-------	-------	-------	-------	----------

Total time = 22.6879s Reduced Chi^2 = 1.1797 Levenberg-Marquardt = 0.01

Free parameter values:

2548	8.419	-2693	404.5	0.00777
------	-------	-------	-------	---------

Iteration = 5

Total time = 27.6245s	Reduced Chi^2 = 2.7318	Levenberg-Marquardt = 0.001
Free parameter values:		
587	4.703 -1557	238.6 0.007769
Total time = 28.6627s	Reduced Chi^2 = 1.1832	Levenberg-Marquardt = 0.01
Free parameter values:		
1959	7.247 -2332	351.9 0.007765
Total time = 29.6794s	Reduced Chi^2 = 1.1765	Levenberg-Marquardt = 0.1
Free parameter values:		
2377	8.031 -2579	387.7 0.00777

Iteration = 6

Total time = 34.6968s	Reduced Chi^2 = 1.1799	Levenberg-Marquardt = 0.01
Free parameter values:		
1809	6.947 -2241	338.3 0.007767
Total time = 35.7618s	Reduced Chi^2 = 1.1734	Levenberg-Marquardt = 0.1
Free parameter values:		
2200	7.694 -2472	372 0.00777

Iteration = 7

Total time = 40.7043s	Reduced Chi^2 = 1.1805	Levenberg-Marquardt = 0.01
Free parameter values:		
1540	6.383 -2068	313.1 0.007758
Total time = 41.6679s	Reduced Chi^2 = 1.1781	Levenberg-Marquardt = 0.1
Free parameter values:		
1992	7.284 -2345	353.4 0.007761
Total time = 42.6747s	Reduced Chi^2 = 1.1774	Levenberg-Marquardt = 1
Free parameter values:		
2135	7.565 -2432	366.1 0.007766
Total time = 43.6199s	Reduced Chi^2 = 1.177	Levenberg-Marquardt = 100
Free parameter values:		
2193	7.681 -2468	371.4 0.00777
Total time = 44.5811s	Reduced Chi^2 = 1.1714	Levenberg-Marquardt = 10000
Free parameter values:		
2199	7.693 -2472	372 0.00777

Iteration = 8

Total time = 49.5144s	Reduced Chi^2 = 1.1759	Levenberg-Marquardt = 1000
Free parameter values:		
2197	7.689 -2471	371.8 0.00777
Total time = 50.5291s	Reduced Chi^2 = 1.1808	Levenberg-Marquardt = 10000
Free parameter values:		
2199	7.692 -2472	371.9 0.00777
Total time = 51.5244s	Reduced Chi^2 = 1.1779	Levenberg-Marquardt = 100000
Free parameter values:		
2199	7.693 -2472	372 0.00777
Total time = 52.5595s	Reduced Chi^2 = 1.178	Levenberg-Marquardt = 10000000
Free parameter values:		
2199	7.693 -2472	372 0.00777
Total time = 53.5818s	Reduced Chi^2 = 1.1745	Levenberg-Marquardt = 1000000000
Free parameter values:		
2199	7.693 -2472	372 0.00777
Total time = 54.6037s	Reduced Chi^2 = 1.175	Levenberg-Marquardt = 100000000000
Free parameter values:		
2199	7.693 -2472	372 0.00777

*** No improvement in chi-squared over previous iteration ***

Total time = 54.6108s	Reduced Chi^2 = 1.1714	
Free parameter values:		
2199	7.693 -2472	372 0.00777

Fit converged:

Parameter values (free parameters with error estimates):

 Foreground parameter values:

1	1		
2	8		
3	2199	+/-	484.9
4	7.693	+/-	0.9488
5	-2472	+/-	360.1
6	372	+/-	31.52
7	0		
8	0		
9	0		
10	0		

 Background parameter values:

1	0.00777	+/-	0.0009677
2	0		
3	0		

 Covariance matrix for free parameters:

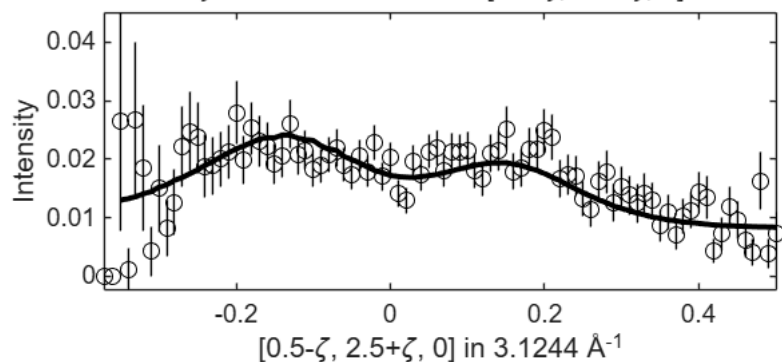
1.0000	0.7127	0.9878	-0.9641	-0.6654
0.7127	1.0000	0.7006	-0.7256	-0.9392
0.9878	0.7006	1.0000	-0.9860	-0.6028
-0.9641	-0.7256	-0.9860	1.0000	0.5924
-0.6654	-0.9392	-0.6028	0.5924	1.0000

Fe_ei800_align_sym3.sqw

S= 7.7±0.95; J0=3.7e+02± 32; J1= 0± 0; gamma=2199.32±484.85

$\xi \leq 0.1$ in $[0.5+\xi, 2.5+\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[0.5, 2.5, -\eta]$, $370 \leq$

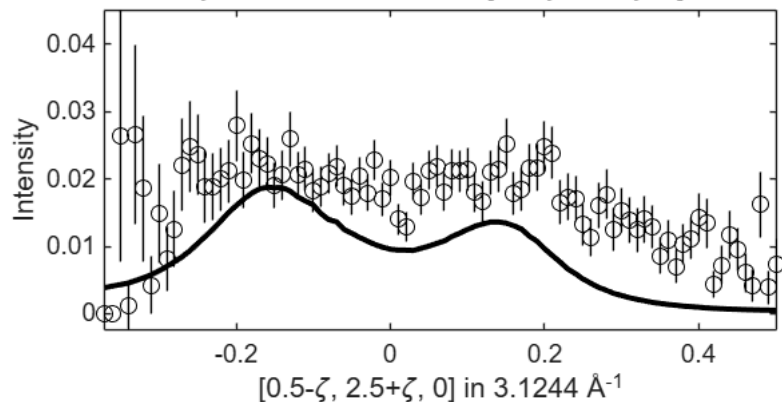
$\zeta = -0.505:0.01:0.505$ in $[0.5-\zeta, 2.5+\zeta, 0]$



Fe_ei800_align_sym3.sqw

$\xi \leq 0.1$ in $[0.5+\xi, 2.5+\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[0.5, 2.5, -\eta]$, $370 \leq$

$\zeta = -0.505:0.01:0.505$ in $[0.5-\zeta, 2.5+\zeta, 0]$



**** En = 385±5 going to 405

Warning: Dataset:2 points with zero, negative or undefined (infinite or NaN) error bars have been removed from fit, which is 1.9802

Beginning fit (max 20 iterations)-----

Starting point

Total time = 0.79076s Reduced Chi^2 = 8.8258

Free parameter values:

2199	7.693	-2472	372	0.016
------	-------	-------	-----	-------

Iteration = 1

Total time = 5.0051s Reduced Chi^2 = 1.0871 Levenberg-Marquardt = 0.1

Free parameter values:

2445	6.352	-2517	371.7	0.009842
------	-------	-------	-------	----------

Iteration = 2

Total time = 9.2976s Reduced Chi^2 = 1.0522 Levenberg-Marquardt = 0.01

Free parameter values:

2930	7.748	-2495	373.3	0.008662
------	-------	-------	-------	----------

Iteration = 3

Total time = 13.5437s Reduced Chi^2 = 1.0531 Levenberg-Marquardt = 0.001

Free parameter values:

3669	9.068	-2528	387.5	0.008205
------	-------	-------	-------	----------

Total time = 14.3727s Reduced Chi^2 = 1.05 Levenberg-Marquardt = 0.01

Free parameter values:

3335	8.536	-2480	377.1	0.008343
------	-------	-------	-------	----------

Iteration = 4

Total time = 18.5637s Reduced Chi^2 = 1.0512 Levenberg-Marquardt = 0.001

Free parameter values:

3435	8.794	-2342	362.6	0.008084
------	-------	-------	-------	----------

Total time = 19.4007s Reduced Chi^2 = 1.05 Levenberg-Marquardt = 0.01

Free parameter values:

3480	8.827	-2421	372.1	0.00817
------	-------	-------	-------	---------

Fit converged:

Parameter values (free parameters with error estimates):

 Foreground parameter values:

1	1		
2	8		
3	3480	+/-	1.145e+05
4	8.827	+/-	159.2
5	-2421	+/-	4.11e+04
6	372.1	+/-	6198
7	0		
8	0		
9	0		
10	0		

 Background parameter values:

1	0.00817	+/-	0.002118
2	0		
3	0		

 Covariance matrix for free parameters:

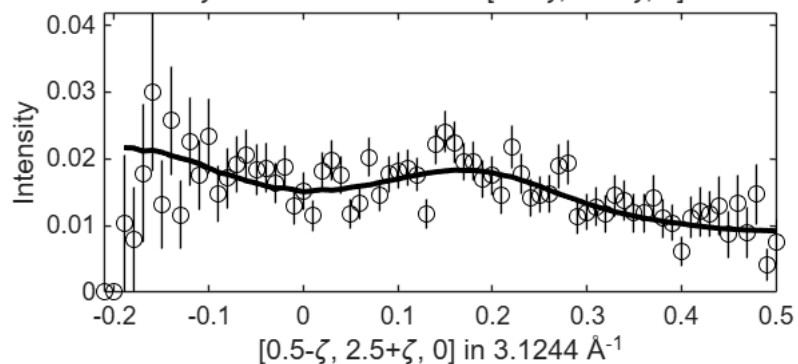
1.0000	1.0000	-0.9997	0.9998	-0.5894
1.0000	1.0000	-0.9996	0.9997	-0.5943
-0.9997	-0.9996	1.0000	-1.0000	0.5723
0.9998	0.9997	-1.0000	1.0000	-0.5765
-0.5894	-0.5943	0.5723	-0.5765	1.0000

Fe_ei800_align_sym3.sqw

8.8±1.6e+02; J0=3.7e+02±6.2e+03; J1= 0± 0; gamma=3480.46±11

$\xi \leq 0.1$ in $[0.5+\xi, 2.5+\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[0.5, 2.5, -\eta]$, $380 \leq$

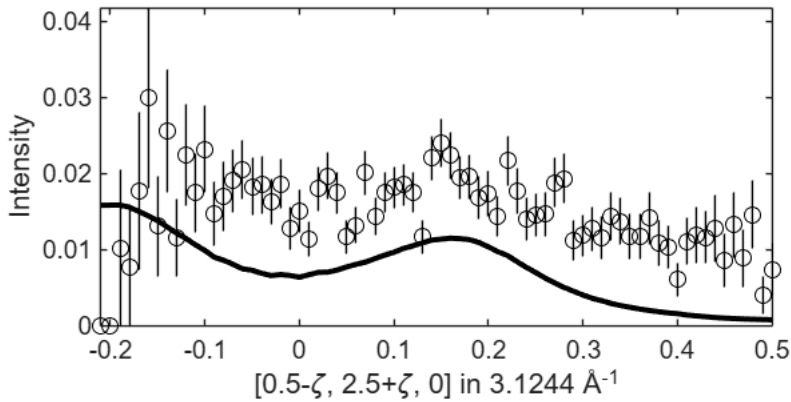
$\zeta = -0.505:0.01:0.505$ in $[0.5-\zeta, 2.5+\zeta, 0]$



Fe_ei800_align_sym3.sqw

$\xi \leq 0.1$ in $[0.5+\xi, 2.5+\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[0.5, 2.5, -\eta]$, $380 \leq$

$\zeta = -0.505:0.01:0.505$ in $[0.5-\zeta, 2.5+\zeta, 0]$



**** En = 395±5 going to 405

Warning: Dataset:1 points with zero, negative or undefined (infinite or NaN) error bars have been removed from fit, which is 0.9901

Beginning fit (max 20 iterations)-----

Starting point

Total time = 0.73567s Reduced Chi^2 = 7.4808

Free parameter values:

3480 8.827 -2421 372.1 0.01512

Iteration = 1

Total time = 4.5163s Reduced Chi^2 = 1.0357 Levenberg-Marquardt = 0.1

Free parameter values:

2573 6.833 -2595 381.7 0.009112

Iteration = 2

Total time = 8.2712s Reduced Chi^2 = 1.0064 Levenberg-Marquardt = 0.01

Free parameter values:

2155 7.376 -2693 381.7 0.008345

Iteration = 3

Total time = 12.1263s Reduced Chi^2 = 1.0054 Levenberg-Marquardt = 0.001

Free parameter values:

2463 7.883 -2950 416 0.008446

Iteration = 4

Total time = 15.8739s Reduced Chi^2 = 0.99849 Levenberg-Marquardt = 0.0001

Free parameter values:

3033 8.897 -3359 471 0.008484

Iteration = 5

Total time = 19.6609s Reduced Chi^2 = 1.3701 Levenberg-Marquardt = 1e-05

Free parameter values:

1.127e+04 22.28 -8106 1125 0.008426

Total time = 20.4039s Reduced Chi^2 = 1.0203 Levenberg-Marquardt = 0.0001

Free parameter values:

5705 13.23 -4937 687 0.008496

Total time = 21.1508s Reduced Chi^2 = 0.99773 Levenberg-Marquardt = 0.001

Free parameter values:

3833 10.18 -3869 539.4 0.008518

Fit converged:

Parameter values (free parameters with error estimates):

Foreground parameter values:

1 1
2 8
3 3833 +/- 9.425e+04
4 10.18 +/- 136.1
5 -3869 +/- 4.855e+04
6 539.4 +/- 6699
7 0
8 0
9 0
10 0

Background parameter values:

1 0.008518 +/- 0.001415
2 0
3 0

Covariance matrix for free parameters:

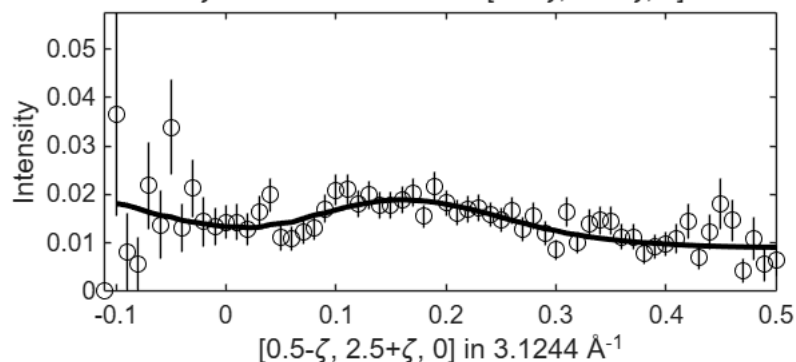
1.0000 1.0000 -0.9997 0.9998 -0.5727
1.0000 1.0000 -0.9996 0.9998 -0.5759
-0.9997 -0.9996 1.0000 -1.0000 0.5560
0.9998 0.9998 -1.0000 1.0000 -0.5592
-0.5727 -0.5759 0.5560 -0.5592 1.0000

Fe_ei800_align_sym3.sqw

10±1.4e+02; J0=5.4e+02±6.7e+03; J1= 0± 0; gamma=3832.83±94

$\xi \leq 0.1$ in $[0.5+\xi, 2.5+\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[0.5, 2.5, -\eta]$, $390 \leq$

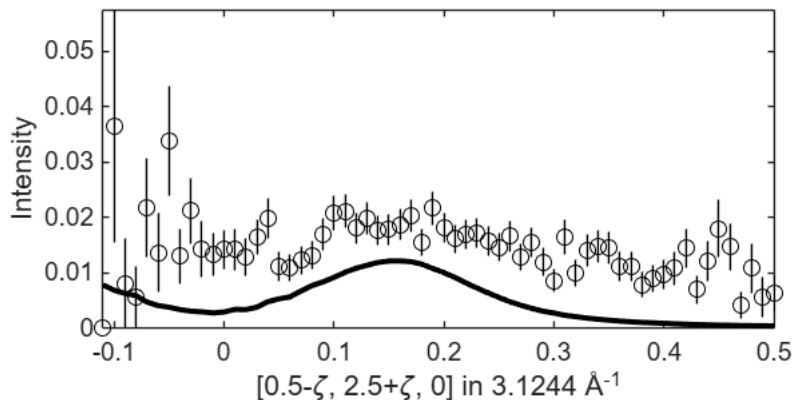
$\zeta = -0.505:0.01:0.505$ in $[0.5-\zeta, 2.5+\zeta, 0]$



Fe_ei800_align_sym3.sqw

$\xi \leq 0.1$ in $[0.5+\xi, 2.5+\xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[0.5, 2.5, -\eta]$, $390 \leq$

$\zeta = -0.505:0.01:0.505$ in $[0.5-\zeta, 2.5+\zeta, 0]$



**** En = 405±5 going to 405

Warning: Dataset:1 points with zero, negative or undefined (infinite or NaN) error bars have been removed from fit, which is 0.9901

Beginning fit (max 20 iterations)-----

Starting point

Total time = 0.62114s	Reduced Chi^2 = 4.9281
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Free parameter values:

3833	10.18	-3869	539.4	0.01316
------	-------	-------	-------	---------

Iteration = 1

Total time = 3.7022s	Reduced Chi^2 = 0.99081	Levenberg-Marquardt = 0.1
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Free parameter values:

6774	10.61	-3655	565.5	0.008647
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Iteration = 2

Total time = 6.7992s	Reduced Chi^2 = 0.99159	Levenberg-Marquardt = 0.01
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Free parameter values:

1.792e+04	19.64	-3072	638.3	0.00657
-----------	-------	-------	-------	---------

Total time = 7.4735s	Reduced Chi^2 = 0.93645	Levenberg-Marquardt = 0.1
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Free parameter values:

1.17e+04	14.18	-3321	583.8	0.007655
----------	-------	-------	-------	----------

Iteration = 3

Total time = 10.5032s	Reduced Chi^2 = 0.94025	Levenberg-Marquardt = 0.01
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Free parameter values:

2.615e+04	24.8	-2573	636.3	0.005772
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Total time = 11.1253s	Reduced Chi^2 = 0.9051	Levenberg-Marquardt = 0.1
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Free parameter values:

1.729e+04	18.27	-3003	592.2	0.006865
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Iteration = 4

Total time = 14.3291s	Reduced Chi^2 = 0.8968	Levenberg-Marquardt = 0.01
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Free parameter values:

3.179e+04	28.89	-2100	605.4	0.005038
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Iteration = 5

Total time = 17.3684s	Reduced Chi^2 = 0.97107	Levenberg-Marquardt = 0.001
Free parameter values:		
7.353e+04	52.72	-1321
Total time = 17.9752s	Reduced Chi^2 = 0.87239	Levenberg-Marquardt = 0.01
Free parameter values:		
4.791e+04	39.06	-1497

Iteration = 6

Total time = 21.105s	Reduced Chi^2 = 0.90203	Levenberg-Marquardt = 0.001
Free parameter values:		
9.663e+04	64.04	-757.2
Total time = 21.7221s	Reduced Chi^2 = 0.86374	Levenberg-Marquardt = 0.01
Free parameter values:		
6.462e+04	48.73	-1049

Iteration = 7

Total time = 24.7897s	Reduced Chi^2 = 0.87634	Levenberg-Marquardt = 0.001
Free parameter values:		
1.185e+05	73.11	-452.2
Total time = 25.4265s	Reduced Chi^2 = 0.86423	Levenberg-Marquardt = 0.01
Free parameter values:		
8.235e+04	57.53	-707.8
Total time = 26.0356s	Reduced Chi^2 = 0.86207	Levenberg-Marquardt = 0.1
Free parameter values:		
7.036e+04	51.67	-965.4

Iteration = 8

Total time = 29.084s	Reduced Chi^2 = 0.86052	Levenberg-Marquardt = 0.01
Free parameter values:		
8.897e+04	60.4	-540.7

Iteration = 9

Total time = 32.2271s	Reduced Chi^2 = 0.87533	Levenberg-Marquardt = 0.001
Free parameter values:		
1.602e+05	87.58	-18.52
Total time = 32.8683s	Reduced Chi^2 = 0.86048	Levenberg-Marquardt = 0.01
Free parameter values:		
1.12e+05	69.61	-257.7

Fit converged:

Parameter values (free parameters with error estimates):

Foreground parameter values:

1	1		
2	8		
3	1.12e+05	+/-	5.738e+08
4	69.61	+/-	1.79e+05
5	-257.7	+/-	7.059e+05
6	714.2	+/-	1.832e+06
7	0		
8	0		
9	0		
10	0		

Background parameter values:

1	0.002061	+/-	0.008012
2	0		
3	0		

Covariance matrix for free parameters:

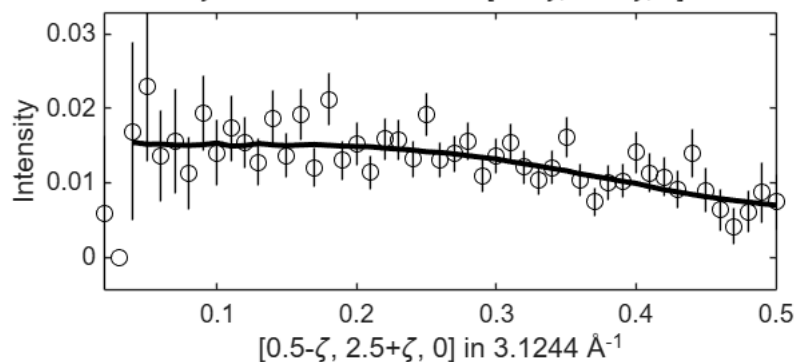
1.0000	1.0000	-1.0000	1.0000	-0.8023
1.0000	1.0000	-1.0000	1.0000	-0.8024
-1.0000	-1.0000	1.0000	-1.0000	0.7985
1.0000	1.0000	-1.0000	1.0000	-0.8022
-0.8023	-0.8024	0.7985	-0.8022	1.0000

Fe_ei800_align_sym3.sqw

$0 \pm 1.8 \times 10^5$; $J_0 = 7.1 \times 10^2 \pm 1.8 \times 10^6$; $J_1 = 0 \pm 0$; $\gamma = 111972 \pm 5.737$

$\xi \leq 0.1$ in $[0.5 + \xi, 2.5 + \xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[0.5, 2.5, -\eta]$, $400 \leq$

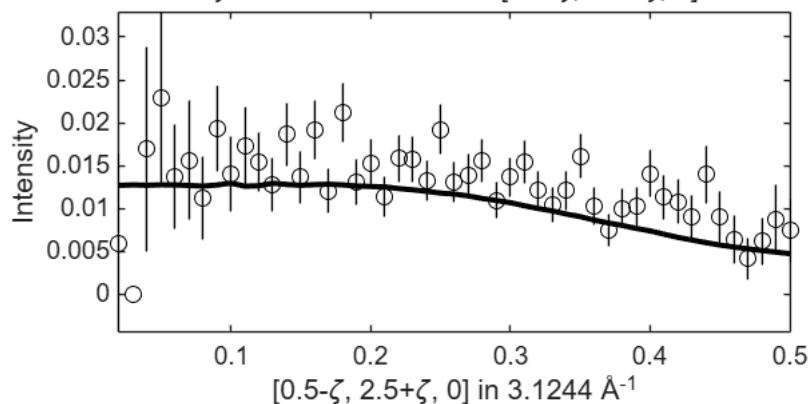
$\zeta = -0.505:0.01:0.505$ in $[0.5 - \zeta, 2.5 + \zeta, 0]$

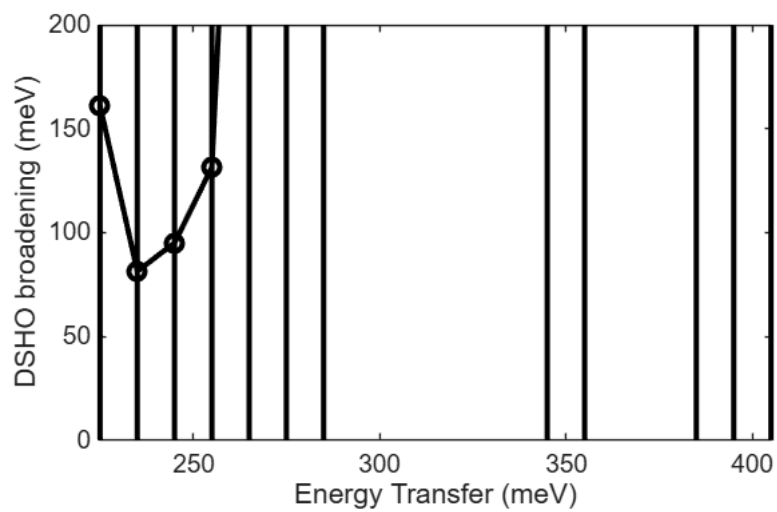
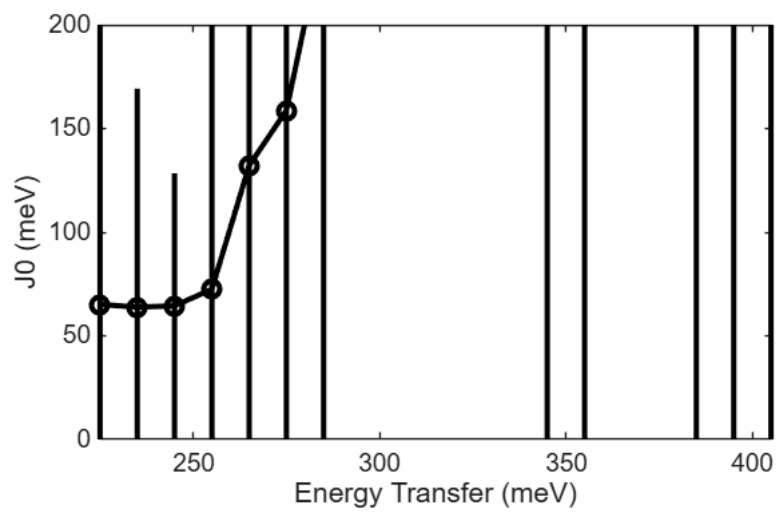
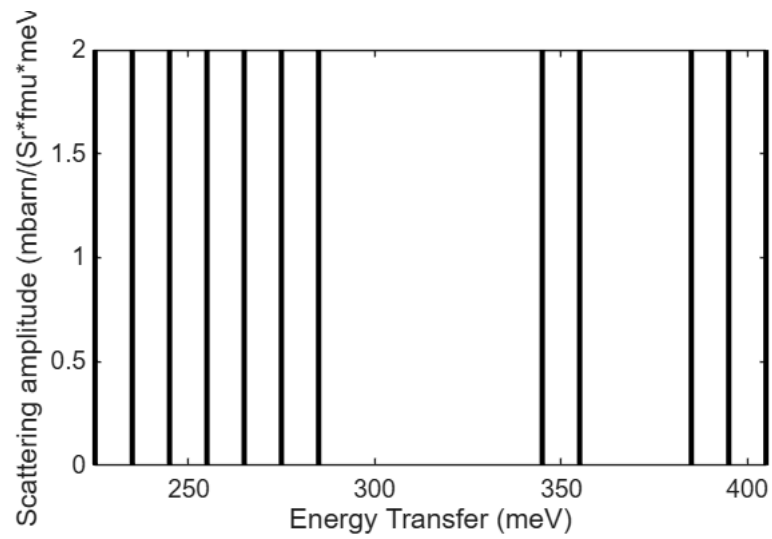


Fe_ei800_align_sym3.sqw

$\xi \leq 0.1$ in $[0.5 + \xi, 2.5 + \xi, 0]$, $-0.1 \leq \eta \leq 0.1$ in $[0.5, 2.5, -\eta]$, $400 \leq$

$\zeta = -0.505:0.01:0.505$ in $[0.5 - \zeta, 2.5 + \zeta, 0]$





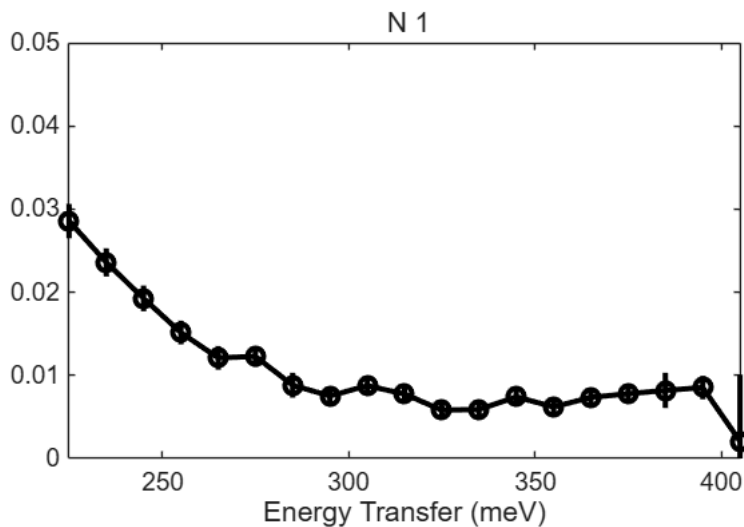
```
c1 = HN_FFCutsFits.(selectCuts);
HN_FFCutsFits.(selectCuts) = [c1{:}];
```

```
beep
beep
beep
```

```

all_fit_par = HN_FFCutsFits.([selectCuts, '_all_fit_par']);
bg_data = extract_bg_par(all_fit_par,[1,2]);
% original slope parameters
pd(bg_data(1));lx(all_fit_par(1).en,all_fit_par(end).en);ly 0 0.05; keep_figure;

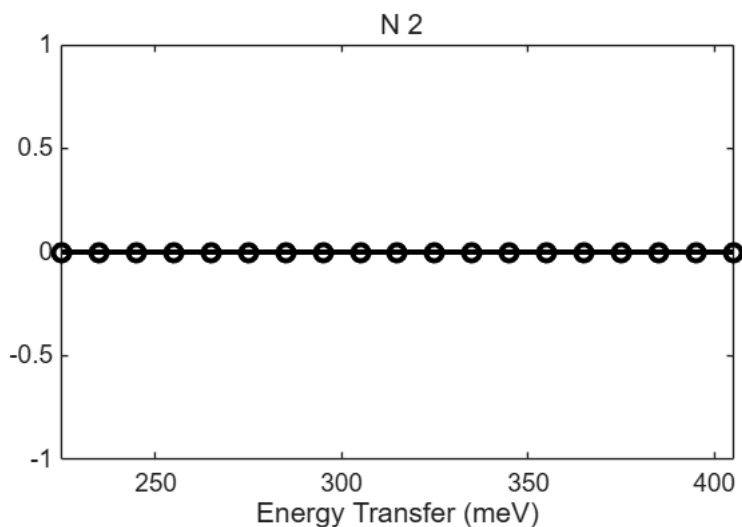
```



```

pd(bg_data(2));lx(all_fit_par(1).en,all_fit_par(end).en);keep_figure

```



```

%pd(bg_data(3));lx(fit_res.peak_fit_par(1).en,fit_res.peak_fit_par(end).en);keep_fi
figure;
%pd(bg_data(4));lx(fit_res.peak_fit_par(1).en,fit_res.peak_fit_par(end).en);keep_fi
figure

```

```

doc = matlab.desktop.editor.getActive;
doc.save;
thisFile = 'e:\SHARE\Fe\Data\analysis\06_fit_with_AQ0\fit_cutFe_HN.m';
targ_file = fullfile(fileparts(thisFile),['fit_cutFe_HN_',selectCuts, '.pdf']);
export(thisFile,targ_file);

```

Error using export (line 19)
Unable to find file.

```
%close all;  
%end
```

```
fit_src_struct = HN_FFcutsFits;  
save(CutFitFileHN,"fit_src_struct",-v7.3')
```